1.1. The project summary

Project Number [1]	870148	Project Acronym [2]	DIY4U

One form per project

General information

Project title [3]	Open Innovation Digital Platform and Fablabs for Collaborative Design and Production of personalised/customised FMCG
Starting date [4]	The first day of the month after the signature by the Commission
Duration in months [5]	36
Call (part) identifier [6]	H2020-NMBP-FOF-2019
Topic	DT-FOF-05-2019 Open Innovation for collaborative production engineering (IA)
Fixed EC Keywords	Open innovation, Process innovation, SME support, Product innovation, New business opportunities
Free keywords	

Abstract [7]

DIY4U aims to develop and promote the adoption of collaborative production engineering approaches in the Fast Moving Consumer Goods (FMCG) sector. The concept is an extension of the format for paint customization at a DIY store where they can make any small quantity of any colour a customer requires. It focuses on particulate and liquid FMCG like detergents, cosmetics, food, etc. To limit the project's scope, we will exemplify DIY4U with detergents and soaps as they are amongst the most complex FMGCs to formulate, represent 84.6 billion € market, and allow us to test DIY4U for both particulates and liquids. A key DIY4U differentiator vs. paint machines is the use of web-based portals to allow users to design their own products from home with a much broader range of performance attributes (than just colour) and then have these produced and delivered in a "designed for you" way. For example, DIY4U will enable a consumer to design and make a personal version of a washing powder with more stain power, or cosmetic (e.g. cream/foundation) that better matches an individual's complexion. The lack of relevant digital infrastructure and efficient small-scale manufacturing facilities limits FMCG product personalisation/customisation. DIY4U will address the blockers of product customisation and small-scale manufacturing by developing an Open Innovation (OI) digital B2B/B2C platform and small-scale automated manufacturing machines (fablabs) for collaborative design and production of personalised/customized FMCG.

1.2. List of Beneficiaries

Project Number [1]	870148	Project Acronym [2]	DIY4U

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	SINTEF AS	SINTEF	Norway	1	36
2	PROCTER & GAMBLE TECHNICAL CENTRES LIMITED	PGUK	United Kingdom	1	36
3	CENTRE FOR PROCESS INNOVATION LIMITED LBG	CPI	United Kingdom	1	36
4	Teknologian tutkimuskeskus VTT Oy	VTT	Finland	1	36
5	FUNDACION CENTRO TECNOLOGICO METALMECANICA Y DEL TRANSPORTE (CETEMET)	CETEMET	Spain	1	36
6	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	IRIS	Spain	1	36
7	DIGITAL CATAPULT	DCC	United Kingdom	1	36
8	INDUSTRIES DU COMMERCE	PICOM	France	1	36
9	CODY AS	Cody AS	Norway	1	36
10	RDIUP	RDI'UP	France	1	36
11	STELAR SECURITY TECHNOLOGY LAW RESEARCH UG	STELAR	Germany	1	36
12	EFFECTIVE DECISIONS SRL	EFF	Romania	1	36
13	DYNAMIC & SECURITY COMPUTATIONS SL	ANALISIS-DSC	Spain	1	36

1. Excellence

DIY4U brings together leading global Fast-Moving Consumer Goods (FMCG) Company and relevant SMEs from across Europe, along with supporting Industry Innovation and Research & Technology Organisations, to accelerate and de-risk the development and use of open innovation and small-scale digital manufacturing approaches for the creation and making of personalized FMCG. i.e. Creating an end to end solution which will enable real time design and delivery of a consumers personalized FMCG product; developing new services and consumer solutions. The project is supported by industry stakeholders, external to the consortium, that have indicated interest to use the technologies and services developed (see letters of support provided in section 4).

The vision of DIY4U is to deliver an end to end digitally supported supply chain which enables personalization of FMCG at home and the point of sale. The project will enhance the innovation capacity and competitiveness of the European FMCG sector through decentralised customer-centric production approaches, making Europe a

leader in the global FMCG industry of the future. This will be achieved by promoting the adoption of Open Innovation (OI) for collaborative production engineering. DIY4U focuses on digitalising and transforming European industries and services involved in the formulation, production and supply of soft matter FMC; across the technical breadth of the industry. The transition from large-scale 'factory' manufacturing via industrial companies of the Do It Yourself (DIY) approaches will stimulate the creation of transformative engineering solutions across the entire European value chain. Fablabs (digitally-enabled small-scale manufacturing machines/factories) have already demonstrated the potential to empower individuals to create smart devices, tailored to local or personal needs, in ways that are unrealizable or uneconomic for mass production.[1] Whilst DIY, fablabs, micro-factories and 'makers' approaches can lead to new functionalities, industry is not yet widely implementing them to engage consumers and drive positive social change e.g. reduced waste.[2] The lack of adequate digital design infrastructure and the inherent inefficiency in making small-batch customized/ personalised products remains a major barrier for European businesses. This means much of the production is outsourced to non-European countries. There is therefore a business opportunity for collaborative companies, especially SMEs, to capitalise on the above new approaches and create cross-industry Open Innovation networks that will enable a wide variety of new customer-centric products and services. Such cross industry interventions can de-risk the adoption of these innovative approaches and avoid the passing on of costs to consumers.

The ambition of DIY4U is to address the blockers of product customisation and small-scale manufacturing and capitalise on the business opportunity by developing an **Open Innovation (OI) digital B2B/B2C platform and fablabs** for collaborative design and small-scale production of personalised or customized soft matter FMCG. **The OI B2B/B2C digital platform** will allow the European FMCG supply chain, including manufacturers, innovation stakeholders (mostly SMEs) and consumers, to seamlessly collaborate on the digital design and digital testing of new personalised or customised powdered/liquid FMCG. To highlight the power of collaboration between disparate stakeholders we plan to decentralize the process by adapting and extending the existing IRIS platforms (IBD® and SHAREBOX) to use distributed ledger technology (blockchain). The proposed platform will use a hybrid network (private permissioned network + a public open network + oracles) combined with secure multi-party computation protocols. The private permissioned network will deal with sensitive data available on top of an access layer, while the public network will hold the data that can be shared between everyone. This approach will open a world of use-cases for integrating the output with digital formulation design tools including such things as soft matter modelling capability, process simulation tools and digital twins, cost and environmental modelling capability, and AI/data analytics capability. **The fablabs (developed and hosted at manufacturing demonstration facilities – MDFs)** will be used for on-demand production of new personalised or customised soft matter FMCG designed using the DIY4U digital platform.

Although DIY and personalization/customization are increasingly being talked about and implemented by some FMCG consumers and innovators (as evident in the multitude of web articles and YouTube Videos[3]), there is a

lack of adequate infrastructures that can enable consumers and new actors to participate in the co-design and comanufacture of customised/personalised soft matter FMCG. Whilst there are strong drivers (e.g. cost-savings, prevention of allergic reactions, preference to smell/colour, preference, health motives, reduced chemical use, environmental protection, etc) for consumers and new FMCG innovators to adopt DIY approaches for product personalisation/customisation, the lack of adequate Open Innovation tools and fablabs have limited adoption to only DIY enthusiasts (DIYers) or high-end consumers who can afford to get their products customised by manufacturers. Soft matter FMCG (e.g. detergents and soaps, cosmetics) produced by DIYers often lack optimal performance and/or fail to meet relevant health and safety requirements. Moreover, the lack of adequate fablabs make current DIY manufacture labour-intensive and inefficient. Some paint manufacturers (e.g. AkzoNobel), supply chain stakeholders (e.g. Riolett Custom Aerosols) and retailers (e.g. B&Q, Halfords) are already offering paint customization services. However, the paint formulation knowhow and specialist manufacturing machines

[1] http://www.imal.org/en/fablab/about

[2] https://ec.europa.eu/research/participants/portal/desktop/en/opportunities/h2020/topics/dt-fof-05-2019.html

[3] https://diyexperience.com/diy-laundry-detergent/; https://kaseytrenum.com/diy-laundry-detergent/, https://homesteading.com/best-homemade-laundry-detergent-recipes-ideas/; https://www.reviewed.com/science/diy-

(requiring trained technicians) are not readily accessible to consumers, entrepreneurs, start-ups and most SMEs. This limits the ability of these stakeholders to rapidly manufacture personalised/customised products for their own use or commercial purposes. Paint services are also limited to colour customisation for a limited product range (broad wall emulsions). A key differentiator of the DIY4U approach vs. the paint machines is the use of webbased portals to allow users to design their own products from home with a much broader range of performance attributes (than just colour) and then have these produced and delivered in a "designed for you" way. For example, DIY4U will enable a consumer to design and collect (in store) a personal version of a washing powder with more stain removal power, or a cosmetic (e.g. face/body cream/foundation) that better matches an individual's complexion or their desired effect.

The proposed DIY4U technologies would be applicable to all soft matter FMCG. However, to limit the scope of the project, DIY4U will use detergents and soaps to exemplify the OI/customer-centric design and DIY manufacturing approach. Detergents and soaps have been chosen as exemplars because: a) Detergents represent some of the most complex formulations in FMCG, both in terms of the number of constituent components and the physical properties of the components. Addressing the challenges of being able to provide tailored formulations within such multi-component systems, while maintaining the overall product stability and performance, will provide the enabling technologies to facilitate adoption of DIY4U by other sectors within the formulated products space. b) Detergents and soaps offer the possibility to test the DIY4U concept on both complex particulate and liquid formulations within a major FMCG market segment; c) The detergent and soap global market is worth ~€84.60 billion[4] and expected to reach €180.55 billion by 2025[5]; d) Consortium partner, P&G, is the leading global manufacturer of detergents and soaps and amongst the 5 largest FMCG companies in the world; e) The consortium's FMCG company (P&G), retail industry innovation hub (PICOM) and innovative SMEs (IRIS, CDY, RDI, DSC, EFF) recognise the trend towards product personalisation/customization and on-demand production and wish to retain leading global market position (P&G) or enhance their competitiveness (P&G and SMEs) by driving the DIY/fablab/makers approaches that are expected to disrupt most industries in the future.

1.1 Objectives

The main objective of DIY4U is to develop and promote the adoption of collaborative production engineering approaches (Open Innovation B2B/B2C digital platform and fablabs) in the FMCG sector. This will spur the rapid production and commercialisation of new/innovative personalised or customised soft matter FMCG for the benefit of consumers. To realise the ambition and main objective of DIY4U, the consortium will work along strategic objectives (SO) measured against quantifiable Key Performance Indicators (KPIs), as follows: Strategic objectives

make-your-own-enzymatic-laundry-detergent; https://www.bing.com/videos/search?q=DIY+detergents&view=detail&mid=842E8E1E1184166993BF842E8E1E1184166993BF&FORM=VIRE; https://www.bing.com/videos/search?q=DIY+detergents&view=detail&mid=0FE6D1F0E8F2B183BC6D0FE6D1F0E8F2B183BC6D&FORM=VIRE

SO1: Development of an open innovation B2B/B2C digital platform: As digitalization is not a goal in itself, the DIY4U digital platform aims to establish and propose clear and transparent links between professionals, consumers and other stakeholders. Open innovation technologies will be deployed and adapted to first design and then realize the digital interfaces. This open access platform will challenge and validate DIY formulations related to the European FMCG sector, focusing initially on soft matter detergent and soap products. The platform will have the capability for industry users/manufacturers to 'plug-in' models for any soft matter products and protect the IP that any model would have. Another objective of this innovative solution is to identify new potential personalized products based on the know-how of creative makers and foster actors to share relevant skills.

[4] https://www.grandviewresearch.com/industry-analysis/soap-detergent-market

[5] https://www.giiresearch.com/report/grvi615535-soap-detergent-market-size-report-by-product.html

KPIs: Feedback from up to 20 SMEs and stakeholders using the platform; >10 new customized products identified.

SO2: Development of a knowledge base (KB) and open innovation technologies: SO2 success will be achieved by:

a) Developing data management plan (DMP) enabling the end to end use of disparate collated data, allowingdifferent actors to interoperate seamlessly.

b) Designing an evolving knowledge base; capable of easily representing information including the needs, userexperience and the data for a given customised formulated product.

c) Developing and integrate a Life Cycle Analysis (LCA) and Parametric Life Cost Calculation Analysis (LCCA) models to support transparent analysis of ingredient and final product features.

d) Plug-in, developing and adapting relevant soft matter modelling and simulation tools to digitally explore the performance of new formulations.

e) Developing and plugging in digital twin of small-scale formulation process machine (fablab)

f) Defining common security and privacy requirements based on European monitoring frameworks to allow thecollection and analyses of private data.

g) Evaluating the proposed solution in terms of the execution requirements.

KPIs: 1 application built on micro services or other API based services, at least 2 applied Machine Learning algorithm, soft matter modelling and simulation tools, LCA/LCCA algorithms, fablab digital twin, performance testing report for all digital components.

SO3: Ensuring secure open data exchange and cybersecurity: This SO aims to secure the data exchange and share skills between designers, makers, users, communities, manufacturers and other actors in an effective and transparent way on the DIY4U platform. Distributed ledger technology (DLT)/blockchain will be deployed to enhance data sharing and transparency, and to track liability and ownership. It is essential to strengthen the ability to detect, mitigate and recover from risks and cyberattacks to enable a secure operational and tactical DIY4U infrastructure.

KPIs: 3partners securely transferring data via the platform, DLT consensus mechanisms auto detecting and responding to cyberattacks attempting to amend records and submit false data.

SO4: Development of Manufacturing Demonstration Facilities (MDF) and DIY4U Training: After analysing user demands and needs, initial small-scale manufacturing trials of the early personalised/customized formulations designed on the platform will be produced and characterized at **MDFs** (STEF and CPI) using existing highthroughput and other complex liquid/particle processing equipment. Learning from these trial runs will feed into the requirements specification and design of fablab machines for automated production of powdered and liquid FMCG. Subsequently, we will demonstrate the feasibility of DIY4U manufacturing by building 2 fablab prototypes (1 powdered and 1 liquid) and using them to provide open access collaborative production. These prototypes will be transitioned to commercial design and replicated after the project's completion. To ensure smooth adoption by consumers, SME users and wider industry, DIY4U platform and fablab training will be developed and delivered to stakeholders including future EU industry workforce via dedicated workshops and apprenticeships at MDFs.

KPIs: 2 Fablab prototypes, Demonstration Reports covering >100 products to check the operability and >10 products to check fablab performance in delivering critical quality attributes. DIY4U Training manuals, At least 4 training sessions delivered to >100 attendees/non-consortium stakeholders, at least 10 students/apprentice trained on DIY4U.

SO5: Empowering different actors to be more involved in the DIY growth and increasing the DIY product acceptance: DIY4U will increase involvement of makers, customers and stakeholders by:

a) Enhancing public and end-user awareness on the DIY creativity aspects by providing modern tools, such as serious games and interactive tools on both the digital platform and fablabs.

b) Identifying and understanding the needs, behaviours and demands of consumers, makers and manufacturers through data collation via cloud-based tools.

c) Running open innovation competitions (OICs) to encourage non-consortium SMEs to build additional functionality atop the DIY4U platform or to develop new innovative/customized FMCG using the DIY4U platform and fablab MDFs.

d) Facilitating the Open Access of the interested stakeholders and widely communicating the positive impacts ofcreativity and DIY4U services and products.

KPIs: 1 fablab application note for customizing products, 1 open serious game, 2 animated video to promote DIY4U, >150 FMCG supply chain companies signed up to DIY4U platform, 2 OICs (>120 SME applicants; 16 Funded); at least 1 non-consortium SME tool integrated into platform, at least 16 new FMCG bespoke products developed by SME via the Open Innovation Projects (OIPs).

SO6: Development of tailored business models to support the sustainability and commercialisation DIY4U outputs: Market development and exploitation of DIY4Y services will be addressed by using agile and demanddriven product development approaches (e.g. Real Time Release). DIY4U will link proven approaches with new strategies to deliver customized products, support the marketing of open innovation/fablab and identify new business opportunities. Market segments, target customers, tailored business strategy and exploitation plan for the transfer of outputs to other FMCG domains and other consumer product manufacturing sectors will be defined. The consortium aims to get the confirmation of the willingness to pay for DIY4U services (platform and fablabs). A post-project business plan and business model for development of the digital platform and fablabs from TRL6 to TRL9 and sustainable DIY4U business operation will be defined.

KPIs: Scalability of the DIY4U platform, number of novel profitable services: Target > 3 smart services, at least 16 SMEs using DIY4U services to manufacture customised products: Target > 200 new long-term and skilled jobs in fablabs and IT sector.

1.2 Relation to the DT-FOF-05 work plan

Fit to the 'Specific Challenge' of the call (Topic identifier: DT-FOF-05-2019)

DIY4U aims to stimulate the transfer from large-scale manufacturing to *industrial companies of the Do It Yourself (DIY), fablabs, micro-factories and makers approaches* in the FMCG sector for powdered and liquid products. The DIY4U approach will lead to *new machines* (fablabs) *and digital services* (OI B2B/B2C digital platform services) that will *facilitate value chain collaborations and consumer involvement* in the design and rapid manufacture of *customer-centric personalised and customized* powdered/liquid FMCG. The DIY4U approach will *reduce time to market* for new personalised/customised FMCG, creating new business opportunities especially for SME and new actors in the powdered/liquid FMCG innovation ecosystem. Demonstrating DIY4U OI digital platform and fablabs with the relevant FMCG supply chain and consumers will raise awareness and confidence in the approach, allowing *industry to widely adopt* these innovative approaches to *engage consumers and respond to societal needs,* taking *individual preferences of women and men into account.*

Fit to the 'Scope' of the call (Topic identifier: DT-FOF-05-2019)

DIY4U focuses on a key consumer-goods sector (FMCG). The OI B2B/B2C digital platform and fablabs will couple design, creativity and knowledge sharing with a customer-driven production approach. The DIY4U approach allows for the *involvement of customers* and the entire value chain in the *co-creation* of products. DIY4U will develop and promote the adoption of collaborative production engineering approaches (OI B2B/B2C digital platform and fablabs) in the FMCG sector. This will spur the rapid production and commercialisation of new/innovative customised FMCG for the benefit of consumers. DIY4U covers the following areas highlighted in scope:

Novel approaches to capitalise on the knowledge and ideas of design and engineering coming from different and even new actors: The Open Innovation B2B/B2C digital platform is a knowledge sharing and knowledge management platform integrated with digital and business process modelling tools (formulation design, formulation modelling and performance simulation and advanced data analytics including cognitive, ontological, social, multi parametric analysis, etc) that allow the whole value chain including consumers and new actors to digitally design and simulate the performance of new formulated powdered/liquid products.

Design of new strategies based on creative and agile methodologies for analysis: Corroboration of formulation performance against simulation data with in-process and final product characterization data from fablabs/MDFs will be used to optimise the formulation models and develop fablab process digital twin. This will progressively enhance the robustness and performance of the digital platform. Real-time analysis of demand and feedback from consumers and visibility of such data across the value chain will facilitate agile responses from the innovation ecosystem in meeting the customers' evolving needs. Consumer feedback will help stakeholders to design high quality 'customized/personalised products. Analytical data tools such as heat mapping and a/b testing will be used for real-time analysis of User Experience and to determine the most effective. Real-time user feedback and

satisfaction ratings on the features of DIY4U platform and fablab will be used to optimise the digital platform and fablab.

Development of open source product data exchange and standard representations of products and processes that ensure the compatibility of modelling and simulation with different process information systems: Information on formulation ingredients/products and processes will be curated and displayed in a consistent digital format, allowing compatibility of modelling and simulation with different process information systems. The DIY4U platform will have the ability to 'plug-in' users' models and protect the related IP. Simulation of new customized/personalised formulated product capabilities will allow validation of compliance to safety and regulatory standards [e.g. EU Detergents Regulation (EC) No 648/2004]. The project will comply with EU General Data Protection Regulation (GDPR) to ensure the security of users' data and use tools developed by project partners to ensure the same. Adequate IP and business models will be implemented to ensure no conflicts arise between manufacturers and suppliers providing components for a 'plug & produce' DIY4U platform and fablab. An open platform based upon a DLT/blockchain-secured data exchange layer will allow manufacturers, suppliers, new FMCG sector actors (e.g. entrepreneurs, SMEs, start-ups) and consumers to collaborate in a way that makes their transactions secure, trustworthy and transparent. For very sensitive information such as prices, partial homomorphic encryption techniques will be used to perform transparent verifications without disclosing the records. Moreover, the deployment of open source tools will promote the open exchange, collaborative participation, and transparency.

Development of new Manufacturing Demonstration Facilities (MDFs), where companies will test new technologies in cooperation with fablabs and makers in order to develop real industrial products and where training is offered: STEF and CPI will host the developed fablabs and act as open access MDFs, offering fablab manufacturing services to individuals and companies (especially SMEs) using the DIY4U digital platform to design new customised/personalised products. The MDFs will also offer training to companies and individuals looking to adopt the new DIY formulation design and manufacturing approach.

Integration of Social Science and Humanities (SSH) aspects regarding creativity: SSH aspects regarding creativity are at the heart of DIY4U as it aims to empower consumers and new actors to participate in the creation and manufacture of new customised/personalised FMCG addressing societal needs. We will analyse the (wide public) human-DIY processes acceptance and define best practices to increase awareness about positive impacts of the DIY products and creativity. Examples of societal drivers for DIY and personalisation/customisation include cost reduction, health reasons, preference for natural ingredients, reduced chemical use and environmental protection. Moreover, legal expertise is involved in the consortium to tackle issues of digital security, data protection and privacy by design.

Facilitating cooperation with other projects: This will be achieved by leveraging partners' involvement in other national and international research and innovation activities linked to DIY4U (section 1.3.5). Cooperation with other projects not involving any consortium partner will be achieved via Outreach activities (section 2.2) such as the EFFRA events.

Enhancing user involvement: via OI digital platform and open access MDFs; upselling and willingness business strategies; creation of an open community & multilingual platform; and royalties for the best shared formulations.

Ensuring the accessibility and reusability of data produced during the project: Defined in Data Management Plan, use of DLT-based records and Research Data Storage Facility (section 2.2a.4)

TRL: Activities will start at TRL4 and achieve TRL6 at the end of the project (section 1.3.3).

Project cost: The project is requesting EC contribution of 5,986,276€ is in the range specified in the scope.

Fit to the 'Expected Impact' of the call (Topic identifier: DT-FOF-05-2019)

Establish Open-Innovation networks for manufacturing that support customer-driven production all around Europe: delivered by DIY4U digital platform and open access fablab MDFs

Creation of specific business models for the engineering of customised solutions, particularly for SMEs, rapid demand changes and shorter time to market: The development of specific business models for DIY4U outputs is a key objective of this project (sections 1.1 and 3.1) and will be delivered in WP2 (section 3).

Improvement of the co-design and co-development capabilities towards a reduction of development costs of new products and services: DIY4U's co-design and co-development digital platform and fablab MDFs for new customized/personalised FMCG will reduce development time/cost and make small- scale manufacture of customized/personalised FMCG economically attractive.

Increase of product variety and personalisation for higher customer satisfaction and loyalty: The digital platform and fablab will allow 'plug and play'/DIY design and rapid manufacture of new personalised/customized formulations meeting individual consumer preferences.

1.3 Concept and approach

1.3.1 Concept

DIY4U concept is to promote the adoption of Open Innovation and DIY/fablab/makers approaches to drive ondemand production of new customer-centric products in the European soft matter FMCG sector. It addresses the key barriers (lack of relevant infrastructures and business models) limiting collaborative small-scale manufacture of personalised/customized FMCG. Our development and trial of an OI B2B/B2C digital platform and fablab MDFs will de-risk the implementation of the DIY4U approach for the co-design and co-manufacture of smallscale personalised/customized FMCG. Figure 1.3a shows a schematic representation of the DIY4U concept including relevant technologies and stakeholders.

The key components of the DIY4U concept are summarised below:

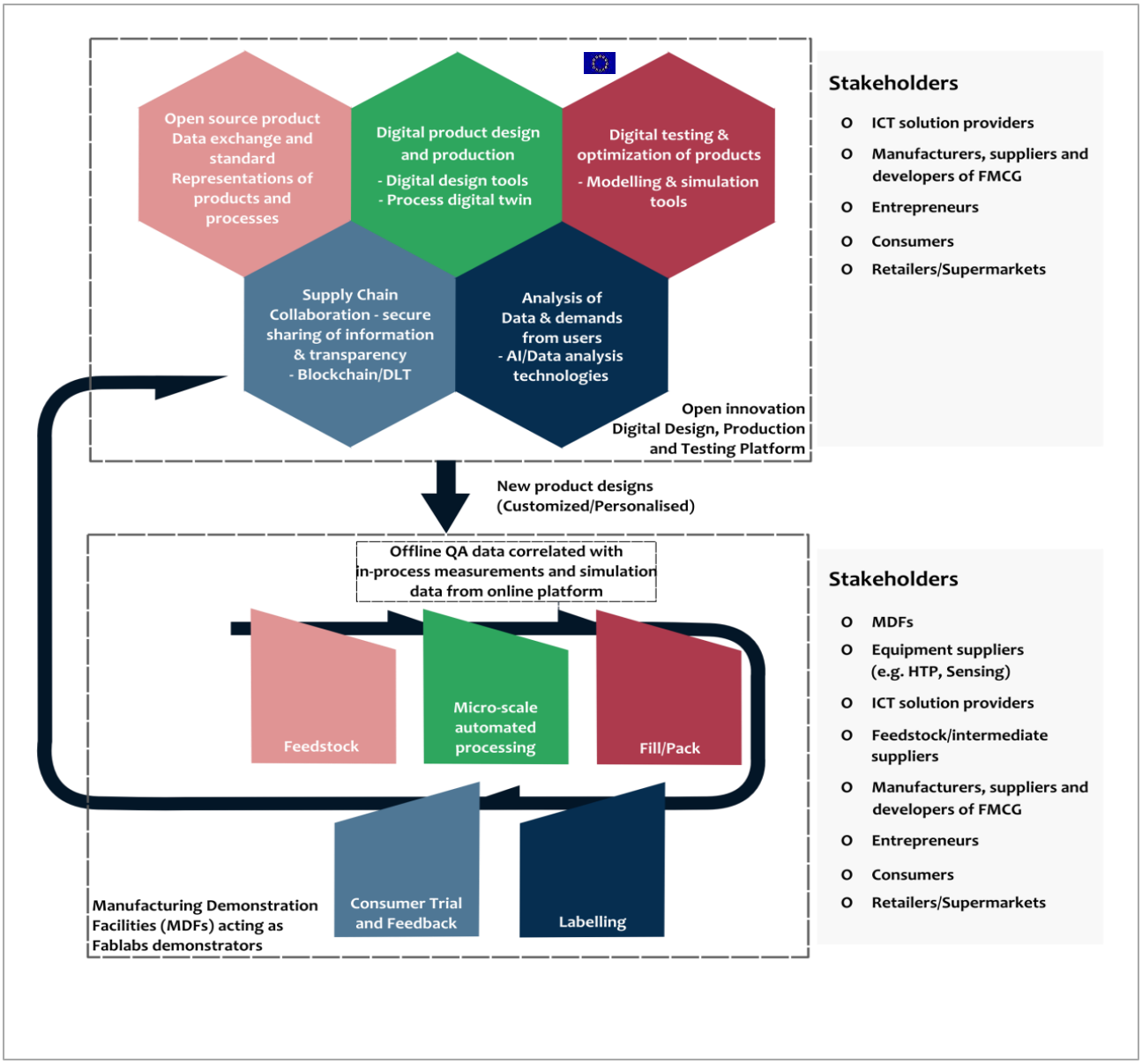

Figure 1.3a Visual representation of the DIY4U concept

The Open Innovation B2B/B2C digital platform will be a highly visible and transparent platform that brings supply chain stakeholders (product & technology developers, ingredient suppliers and manufacturers) and consumers together and enables them to seamlessly collaborate on the efficient creation of new particulate/liquid

FMCG. Integrated digital tools across the product/process/supply chain (molecular modelling & simulation, process simulation & digital twin, cost and environmental modelling, and AI/data analytics) will be used in conjunction with product and process information from ingredient suppliers and manufacturers to digitally design high performance new customised/personalised products. The digital platform will allow all actors (including consumers) to propose and design new products. Tracking individual/corporate IP and

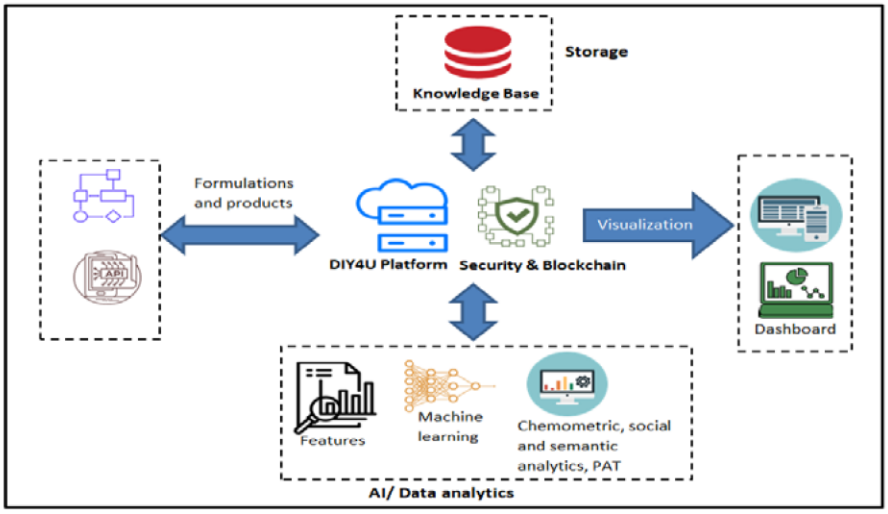

Figure 1.3b DIY4U Platform architecture interactions between partners on the platform will be possible through integration of Enterprise Resource Planning (ERP) and distributed ledger technology (DLT)/blockchain. A consistent integrated data structure will facilitate open source product data exchange and standard representations of products and processes from design to consumer. Blockchain allows digital information (e.g. product/processes information including information on standards and regulatory compliance) to be distributed, shared, persistent and immutable. The ledger will ensure that rights and permissions are respected, allowing industry stakeholders and consumers to co-design and codevelop new customised/personalised products. It is important to note that the DLT/blockchain platform envisaged will not use the same energy-intensive consensus process of bitcoin or other cryptocurrencies (mining) and will instead use an efficient Lamport-type consensus algorithm such as PBFT.

Through engagement with the simulation tools and the serious game, consumers and users will benefit by being involved in the design and manufacturing process and seeing how their formulation choices will impact product performance, environmental and social aspects. Blockchain will also enable the sharing of personalised formulations.

Modelling & simulation tools (e.g. Computational Fluid Dynamics - CFD, LCC/LCCA models, and process models) will be developed as modules to be plugged into the platform to enable users to simulate/optimize the performance and impact of the created formulations prior to making them and to device the optimal processing route.

Big data technologies, integrated in the DIY4U digital platform, will be used to track real-time consumer demand changes and to generate optimised formulations for the consumer. While there is some societal push-back against using people's data, it is logical that such information is employed (whilst complying with GDPR) to provide products of improved performance to the consumer, should they wish. For a laundry detergent exemplar (for instance), big data could leverage the following information to optimise the formulation.

- Family composition, ages of any children, occupation of parents to predict the cleaning requirements of the family and hence deliver an optimised surfactant package.
- Shopping habits, the material and colour of fabrics purchased by the consumer to tailor the formulation towards optimised cleaning, reducing fabric wear and minimising colour fade.
- The fragrance profiles of the products (e.g. deodorants & personal care purchased) purchased to customise the perfume additions to the detergent to synergise with these.
- The location of the consumer's home address such that the 'builder' package of the detergent can be optimised to the water profile of the consumer's tap water, i.e. soft vs hard water.
- Time of the year, to decide how to alter the formulation. For example, stain profiles will change regionally and could be more tailored towards removing grass stains in the summer, while being tailored towards mud in the winter. Moreover, the formulation could be customised based on the colour palette of the clothes worn by the consumer as a function of the time of year, e.g. lighter clothes in the summer and darker clothes in the winter.

Through the methods/technologies integrated, the platform will promote social inclusion and diversity by empowering all consumers/users, irrespective of age, socioeconomic status or gender, to contribute in the codesign of innovative products meeting their needs. This will ultimately increase product variety and personalisation and deliver higher customer satisfaction and loyalty.

Data analysis: The proposed concept for data analysis will improve both the customized formulation and the process in order to define an optimal formulation by taking into account all parameters (costs, time, LC, QbD, colour, composition, health, process …). This will ensure a real-time interaction with end-users to predict their needs, demands, perceptions and behaviours. Figure 1.3c shows the different parts of the DIY4U data analysis concept.
The data analysis process aims to capitalise collaborative Knowledge Base. To reach goal, DIY4U proposes a specific methodology (see figure 1.3c). The elaborated KB will be updated only when a new formulation is created, rejected or improved. The end-users can share knowledge or solicit the DIY4U platform to evaluate a novel formulation.

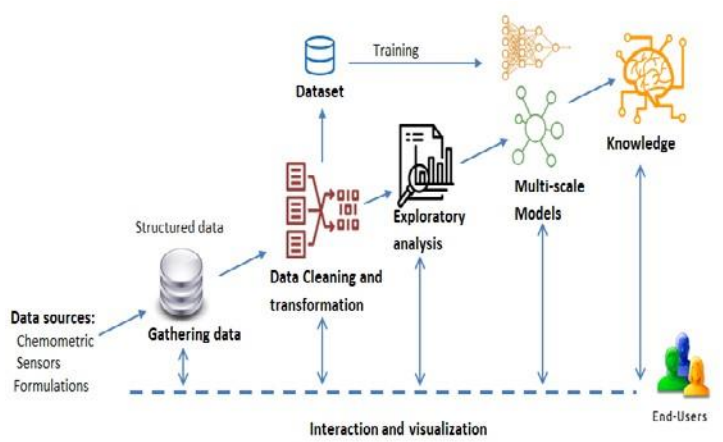

Figure 1.3c Data analysis process

DIY4U Platform Graphical User Interface (GUI): The DIY4U project will develop userfriendly GUI interfaces based on consistent and intuitive design. Visual and functional consistencies are one of the strongest contributors to enhance the usability and learnability. This project will define methods to measure the user satisfaction indicator to improve the user experience through iterative design and usability measurements. The DIY4U platform

will ensure user interaction with information by manipulating efficiently visual widgets and respecting the following principles:

- Reduce the number of actions from the user to achieve the task;
- Ensure that users never lose their information;
- Use clear and meaningful words that help users understand the concept more easily;
- Assure the accessibility and the visibility of the system status;
- Make the platform more comfortable for a user to interact with a product; - Take into account the perception of users.
- Use of smart tools simulations for creation of formulations and awareness assessment.

DIY4U Platform Scalability: The platform's scalability is a main goal for this project. To ensure this important property, the DIY4U platform will deploy open source tools, implement evolutive schema database and apply design patterns principles and the best practices. Moreover, to guarantee the integrity of new industrial domains, the DIY4U project will integrate scalability at several layers:

- Performance: At the beginning, the DIY4U project will ensure a high performance to support many new users and the need to add more capacities.
- Availability (CAP[6]): the DIY4U platform's data layer will be based upon DLT/blockchain which automatically ensures consistency and redundancy across multiple distributed replicas in order to ensure that clients have uninterrupted service. Data stores for larger 'off-chain' digital assets will also be replicated across multiple sites to ensure local availability and redundancy for loss-prevention.
- Costs: Open source solutions will help identify community members that will make great progress. This will reduce the total cost of development and facilitate growth of applications.

Furthermore, DIY4U will, early on, take into account the extensibility of the platform by identifying similarities between domains and generalize as possible the GUI interfaces.

How DIY4U platform would work (figure 1.3d)

Step 1 Sign up: Register for an individual or company account on the DIY4U platform. Watch the virtual tours or tutorials to become ready to formulate in the cloud.

Step 2 Create, test & recommend: Use the DIY4U platform to create customised or personalised candidate formulations, digitally assess the performance of the created formulation (e.g. using plugged-in models incorporating CFD) and simulate the manufacturing of the formulation using the processing digital twin. DIY4U algorithms will automatically compute users' desired observables and recommend the best formulation as well as the most efficient and accurate processing approach.

Step 3 send results to fablab MDF: With a single click, share results of users' optimal formulation and optimal processing conditions with DIY4U Fablab/MDF. Then, the users have the possibility to create another formulation or to quit after filling the satisfaction form. The DIY4U platform will propose a specific interface for manufacturers/makers to receive orders and exchange information with users.

[6] CAP theorem

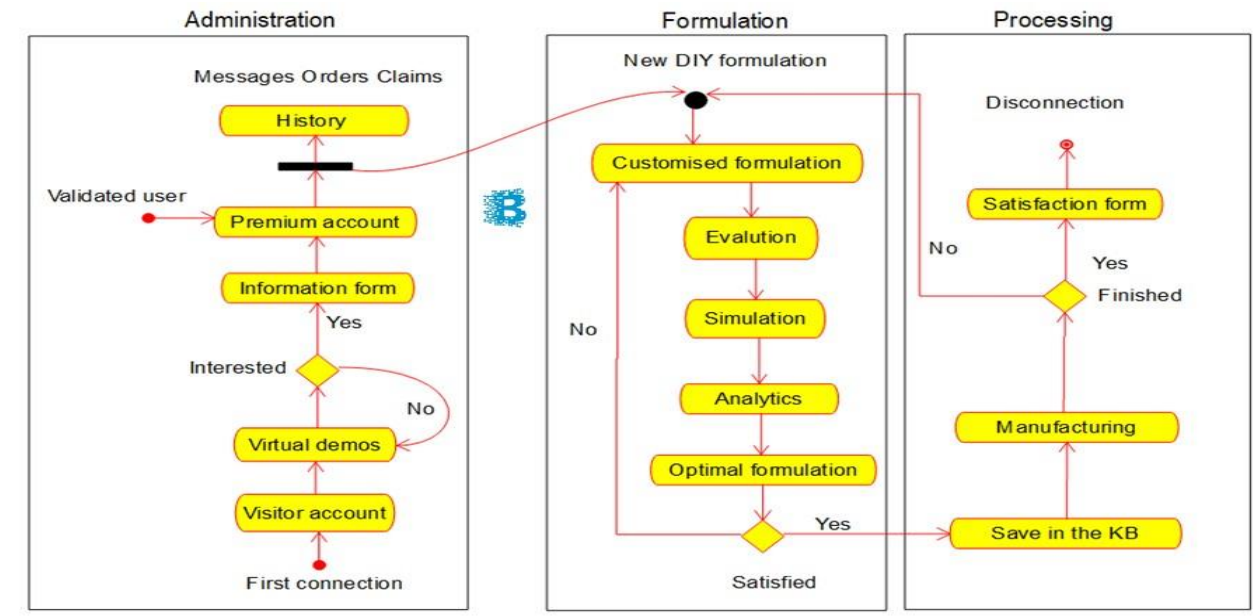

Figure 1.3d Functionalities of the DIY4U platform for users

The Manufacturing Demonstration Facilities (MDFs) developed at STEF and CPI will act as fablab demonstrators, offering open access facilities and resources for entrepreneurs, SMEs and anyone to rapidly and cheaply make personalised/customized soft matter FMCG. The MDFs will be furnished with small-scale digitallyenabled machine (fablab) for rapid manufacture and high speed QA testing of innovative products designed on the DIY4U platform. The fablab will be integrated with high speed labelling capabilities and will address the inefficiency barriers of small batch/customized products (i.e. process efficiency, inventory costs and changeover/downtime). The MDFs will also provide **DIY4U training** to industry stakeholders and individuals willing to trial the DIY4U approach to inform investment decisions (businesses) or foster knowledge (students).

Visualisation method would also be applied in the design of facilities that would be purported to house the fablab; enabling distribution groups to see how the fablabs would incorporate into their businesses and consumer flow patterns with minimum disruption and maximal usage. Quality visualisation will also enable fast efficient staff training, in such areas as materials refill, to enable safe, effective products without the need of skilled formulators at site constantly. This will safe effective upskilling of a broader workforce with the potential for offsite real-time monitoring. The coalition of these applications will mean that efficient customisation and small-scale manufacturing can be applied across product ranges and differing applications.

Figure 1.3e shows a conceptual diagram of a fablab for manufacture of powdered laundry detergents. Feedstocks for creating the personalised formulations are all either powder feedstocks or pre-formed powder composites. For example, the ‘top level’ surfactant package that defines the cleaning potential of the detergent have been preblended into typical filler compounds such as sulphates in order to aid with the flowability to facilitate easy incorporation into the detergent. In the same manner, components that are traditionally added in small quantities to the formulation such as enzymes are not added in their pure form and are instead pre-blended into a baseline powder. This approach affords the platform a number of advantages in the case of powdered laundry detergents: Firstly, it increase the precision to which the small additions can be added to the formulation, facilitating the creation of small quantities for the consumer; secondly, it negates (to a large extent) issues relating to powder blend mixing and

hence product homogeneity leading to better, more consistent products generated in quicker time; finally, it helps to constrain the overall formulation space provided by the fablab, preventing the consumer creating non-viable or poorly performing products. It is logical for detergent manufacturers like P&G to bridge the supply chain by providing such bulk intermediate powders, especially because it is envisaged that the fablab approach will facilitate the uptake of higher value component additions which can be difficult to incorporate in to non-personalised/customised detergents due to market cost drivers. This allows innovative SMEs developing such higher value components to be directly involved in the collaborative manufacturing approach.

A powder processing framework might be required at the back. Such a framework needs to be incorporated at the

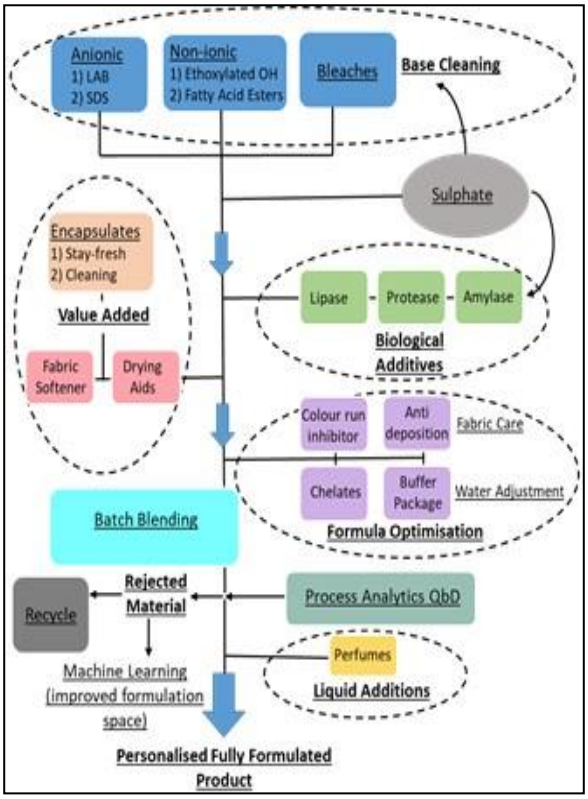

Figure 1.3e DIY4U Fablab concept as applied to powdered laundry detergents

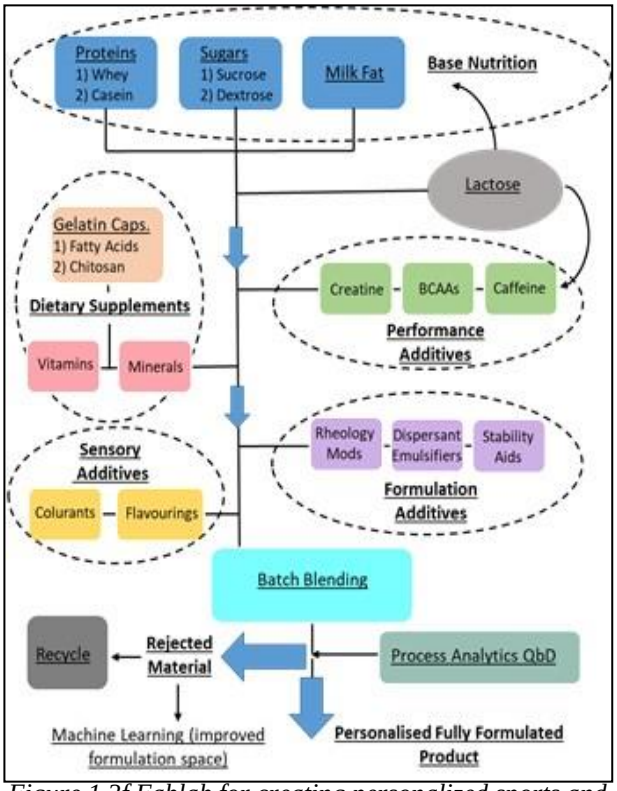

Figure 1.3f Fablab for creating personalized sports and nutrition drinks

point of sale and be sufficiently attractive to the retailer or SME manufacturer who owns the floor space where the fablab will be placed. Moreover, it will be necessary to ensure that the replenishing of the constituent powders can be accomplished using existing staff and infrastructure and without requiring addition of any engineering controls or containment to handle powder transfer.

The powdered detergent fablab exemplar has been used to demonstrate the concept of fablab as opposed to serving a realistic demonstration of the formulation space for a powder laundry detergent. In order to showcase how the approach is universal to multiple FMCG, the same approach has been demonstrated in figure 1.3f for manufacturing of gym supplements at a sports facility or gymnasium. In this iteration, the fablab would be creating customised macro and micro-nutrient sports drinks for users of the gym which they could then form into a nutritional beverage via the addition of milk or water. It can be seen from inspection of figure 1.3f that the approach is very similar to that of a powdered laundry detergent

in figure 1.3e. Indeed, there is nothing to prevent the fablab being set up so that the final liquid addition could be carried out within the device and hence dispense the result as a fully formulated liquid product.

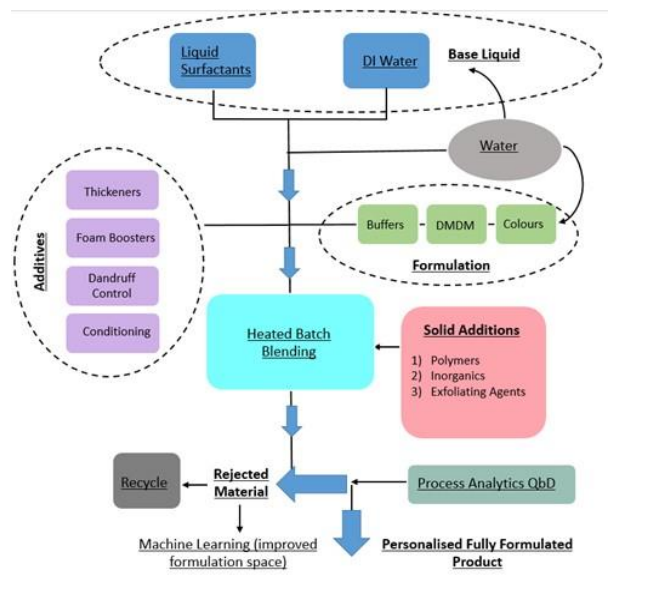

Figure 1.3g: Fablab for making personalized shampoos

Figure 1.3g illustrates a fully liquid-based fablab for manufacture of shampoo, in this case the major processing challenge relates to the rheology of the components which must be blended together and being able to achieve this is acceptable timeframe will be challenging. Hence, rapid heating of the components will be required. Cooling will be achieved by the flow of formulated product through non-heated parts of the dispensing system, with it actually being favourable from a consumer experience perspective for a warm product to be dispensed. Formulation is at the heart of the fablab and hence it is important to ensure 'flexible robustness' in terms of the options available to the consumer. It is important to ensure that consumers are not able to create non-viable, poorly performing, or even dangerous formulations, while at the same time still allowing them to make choices that have a meaningful impact on the product that they create. Furthermore, the fablab needs to provide assurance that the quality of product dispensed meets the requirements of the consumer who has formulated it. *shampoos*

For example, if the consumer specifically requests a 'non-biological' detergent, the fablab will need to check this using in-line process analytics (e.g. Raman, UV, near-IR). Coupled with the real-time process analytics approach for product analysis, the fablab will be integrated with a decision-making process such that sub-optimal formulations or nonviable formulations will not reach the consumer and will hence be recycled via blending into lower value,

non-personalised products. While it will result in a bad user experience for the consumer to order a non-viable product, it is envisaged that by employing self-learning neural networks (on both the digital platform and fablab), DIY4U will quickly be able to understand and prevent such events occurring. Over time, the device, complemented by improvements in the digital design modelling and simulation tools, will work to tighten the formulation space available to the consumer to prevent such bad experiences.

Despite the overall approach of fablab being broadly equivalent for very different consumer goods, the challenges associated with maintaining product quality are subtly different and hence the approach needs to be sufficiently robust to meet these challenges. In the case of the gym supplement exemplar (figure 1.3f), the cleaning of the device will be more prominent with cross contamination between users being important. In this case the process analytics being employed can be used to verify device cleaning and ensure material that exhibits unwanted ingredients above trace levels are recycled into non-personalised products. Provision of flexible PAT ports to allow plugging or replacement of miniaturised PAT tools should mitigate this challenge. Examples of PAT tools that could be integrated include: In-line spectroscopy (Raman, UV, near-IR), Focused beam reflectance method, In-line rheometry/viscometry,

Acoustic emissions sensing, Electrical resistance tomography, Particle size & shape (PVM), Colour analysis, Terahertz imaging, Laser Induced Breakdown Spectroscopy, Turbidity sensing, Hyperspectral Imaging).

Business model Development: PICOM will work together with the consortium and the DIY4U ecosystem engaged to develop a business model that will ensure the industrial adoption, commercialisation and long-term sustainability of the DIY4U platform and fablab.

1.3.2 Methodology

DIY4U will largely focus on developing and demonstrating the Open Innovation B2B/B2C digital platform and fablab MDFs in the detergent and soap domains. To achieve this, we will develop, modify, integrate and test several key components of DIY4U:

Digital platform

- Formulation 'plug and produce'
- Knowledge management and collaborative platform
- Advanced data analytics
- Formulation computational modelling
- Formulation process digital twin
- LCA and cost modelling
- Distributed Ledger Technology/blockchain to track IP and interactions
- Targeted marketing campaigns and open innovation competition to recruit users and suppliers

Fablab MDFs

- Hardware (unit operations)
- Process analytical technology (PAT) and process control algorithms ● Digital twin (digitalisation of formulation manufacturing)

Each of these components are further elaborated upon in this section.

Formulation 'Plug & Produce': Relevant product and process information will be collated from FMCG manufacturers and suppliers and presented to platform users in a consistent format that allows the design of new viable/safe customised formulations via a "plug and produce" approach. Relevant modelling and simulation tools, developed and/or plugged into the DIY4U platform, will allow users to simulate and optimise the performance of the created formulations.

Knowledge management and collaborative (KMC) platform: The KMC platform will facilitate a logical workflow for enabling end-users (including consumers, personalized product providers and Fablab owners) to collaborate throughout the product design, testing, production and delivery life cycle by developing novel product designs and formulations or adapting existing ones. To achieve this, the KMC will provide users with a range of design methodologies to facilitate the process and product design and its subsequent testing, control and optimization. Computational modelling and simulation tools are an important component of the platform, and they will be based on a determination of not only the process drivers (the physical and chemical requirements of the process), but importantly they will be driven by customer satisfaction criteria as well as environmental and business requirements.

The initial Platform Architecture will be validated and adapted based on the House of Quality from the QFD. The methodology that will be used to develop and code the Platform itself will be a proven agile methodologies of SCRUM (that is ideally suited to software developments) and Test Driven Development for efficiently developing "almost-bug-free" large software platforms like IBD® because it is a programming methodology oriented to code "reliable-by-design" software. The Scrum framework operates as follows: A product owner (i.e. represented by DCC with the support of EFF and IRIS) creates

a prioritized wish list called a product backlog (this will be extracted from the House of Quality above). During sprint planning, the Scrum team (the IRIS and EFF software developers) pulls a small chunk from the top of that wish list, a sprint backlog, and decides how to implement those pieces. The team has a certain amount of time — a sprint (usually two to four weeks) — to complete its work, but it meets each day to assess its progress (daily Scrum). Along the way, the Scrum Master (IRIS) keeps the team focused on its goal. At the end of the sprint, the work should be potentially 'shippable': ready to hand to a customer, put on a store shelf, or show to a stakeholder. The sprint ends with a sprint review and retrospective. As the next sprint begins, the team chooses another chunk of the product backlog and begins working again. Scrum is a highly powerful methodology for keeping the end-user engaged and with influence over the development process.

The KMC Platform, which will be deployed through a supercomputer cloud to guarantee the high processing power and speeds required to run the tools, unites expert designed formulation design methodologies, numerical and process designer tools, Digital Twins, LCA and LCC tools, AI analytics, and Blockchain, considering the whole product design cycle, from ideation to fabrication and delivery. The KMC Platform will ensure the following features and performance in terms of simulation functionality:

- The GUI will be user friendly and will allow the potential users feeding the databases, user materials' implementation, etc. by means of custom encapsulated scripts.
- Built-in models will be included in the global process interface in order to avoid the full design of every component (if something exists and it is successful, it should not be designed twice).
- 'One-stop-shop' philosophy. The tool will account for everything involved in the process: from early design, to digital testing and functional simulations; from full product design and analysis, to LCA and environmental impacts evaluation. Integration with relevant databases engineering, physical and chemical, environmental, and economic
- An optimization module, where almost everything should be input and output at a time.
- A representative case study will serve as a preliminary validation example.
- A platform template will be provided as a starting point.
- The global interface will be able to call other software (either commercial or open source) for the particular analyses and process' bottleneck findings.

The methodologies and tools that will be hosted into the Platform are:

Advanced big data analytics methods: There are several open source methods and tools for advanced data analytics. In this project, the following techniques will be deployed and applied:

- DIY4U will define a clustering technique based on a smart mix of algorithms[7] (K-means and OPTICS). This technique will allow for example to segment consumers and classify different products and evaluate the reachability of a new formulation on related cluster.
- Machine learning: In order to analyse and make decision, Decision trees will be adapted and used to assess the relevance of formulation and predict, for example, the behaviours and the demands of user.

Formulation computational modelling: The usage of formulation computational techniques will allow the consumer to create viable, performing, or non-dangerous formulations that would fit with their desire levels of product that the consumer is searching for. The key idea is the development of a series of numerical methods that will forecast how the formulation will perform in the consumer home reducing the testing procedures and not creating a bad experience to the consumer. Soft matter modelling and

[7] Kiran Jyoti "Data Clustering Approach to Industrial Process Monitoring, Fault Detection and Isolation" International Journal of Computer Applications, 2011.

simulation methods (e.g. Langevin dynamics, dissipative particle dynamics, multi-particle collision dynamics/stochastic rotation dynamics, and the lattice-Boltzmann method) will be applied to resolve the multi-scale nature of new FMCG formulations and provide quantitative information to predict their performance. Manufacturers (e.g. P&G) will also be capable to plug-in specific models that allow consumers to create, digitally test and optimise their customised formulations. DIY4U DLT platform will ensure the manufacturer's IP for the model is protected whilst giving access to such models to non-formulation experts and stakeholders.

Formulation process digital twin: CETEMET and DSC will develop a multiphysics, multiscale, probabilistic simulation (digital twin) of the fablab processing equipment (to be developed in the project) using the best physical models, sensor/PAT updates data driven analytical algorithms, partners available knowledge (P&G, CDY, STEF and CPI) and operation history. The formulation process digital twin will virtually represent each of the production processes to obtain a formulation virtually. This virtualization will take advantage of computer modelling that recreates each of the unit operations to mimic the overall equipment performance. The equipment process virtualization will take advantage of Computational Fluid Dynamic (CFD) tools to analyse and optimize each unit operation virtually.

LCA and LCCA Model Sub-platform: DIY4U's LCA/LCCA model sub-platform will be designed to calculate the many permutations given for each material selection and decision point made by the User, and this will give an output result including environmental performance (equivalent CO2/kg) and monetary cost (£/kg). A mass and energy balance approach will be taken to simulate process stream mass flow rates (kg/h) and component concentrations within each stream. A comprehensive database will be created from publicly available data to assign to mass flows, and this will give manufacturing cost estimation and embedded carbon data for each material used in formulation. A decision tree will be built to simulate a User Interface, offering a range of preferences that pre-determine a defined process schedule. Life Cycle Assessment Guidelines ISO 14040/14044 will be adopted to develop the model with a systems-based approach. Life Cycle Costing will allow the identification and evaluation of operational and capital costs (OPEX and CAPEX), and output results will include monetary product values, as well as other economic figures such as Net Present Value and Return of Investment. The LCC assessment is performed in parallel to the LCA, making the same assumptions and using the same functional unit.

Distributed Ledger Technology/Blockchain: A hybrid blockchain network (private permissioned DLT + public open network) will be deployed as the 'base messaging layer' between parties and will enforce a consistent structure of corporate or agent identity, process, mixture and ingredient metadata, as well as time-stamping and sequencing of requests and responses between parties to the ledger from different fablab sites, contributing companies, resellers or even consumers, and the use of modules atop the platform (e.g. CFD or analytics). There will also be the potential for process automation through 'smart contracts', such as preventing dangerous mixtures or use of stale batches, or only allowing certain parties access to particular IP. Cryptography (including zeroknowledge proofs) will be used to ensure proprietary information is only legible by authorised parties, thus

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increasing the trust from corporate partners and increasing their willingness to adopt the platform. Consistent and reliable messaging will ensure preservation and attribution of intellectual property developed on the platform, and uniformity of the data will allow it to be fed into machine learning processes developed for the platform. This will also enable a 'reporting module' to extract data from the ledger if requested for legal compliance processes. The DLT will be designed with open innovation in mind, and have well-documented APIs that will integrate with existing ERP systems and other graphical front-ends. It will also be possible to extend the system with automated payment methods beyond the end of this project, once there is more regulatory clarity.

To comply with GDPR there will be no recording of PII onto the ledger, merely off-chain look-up locations or hashes to external databases (which could also be distributed between the parties using IPFS technology) which comply with the right to deletion.

Fablab MDFs: CPI and STEF will host fablabs developed and provide open access for fablab demonstration and DIY4U design and manufacturing training.

Fablab Design: The consortium will engage potential fablab users and supply chain stakeholders to define user requirement specifications that will be used to develop technical specification and a design package for the fablabs. Key stakeholders to be engaged in the specification and design of the fablab will include: FMCG consumers engaged through a user research carried out on the DIY4U digital platform, FMCG SME innovators and retailers signed up from PICOM's retail innovation cluster and via user research on the DIY4U platform, consortium FMCG manufacturer (P&G), FMCG ingredient suppliers engaged via P&G's network and through targeted user research campaigns on the DIY4U platform, robotic and automation solution providers/equipment manufacturers (e.g. consortium partners – CDY, CTM, VTT, DSC), and consortium MDF partners (CPI and STEF). The involvement of these stakeholders and potential users (from across the FMCG development and manufacturing innovation continuum and supply chain) in the specification and design of the fablab will ensure maximum impact of the fablab MDFs on all stakeholders and target market segments.

Powder Detergent manufacturing fablab design considerations: The main mixing processes for manufacturing detergent powders (spray drying, agglomeration and dry blending) will be reviewed and considered together with other user requirement specifications to select the most appropriate processing method or combination of methods to be integrated into the fablab. One intent of **spray drying** is to convert liquid materials such as (polymers/surfactants/silicates) into a powder form by mixing them with compatible powders (sulphate/carbonate for example) to create a "base powder" from which the final detergent is built. Since spray drying is complex and costly, we do not expect to integrate it into the fablab. We expect detergent manufacturers like P&G to supply base powders as feedstock for the fablab. **Agglomeration** involves blending of dry raw materials with liquid ingredients using machines such as liquid binder, rolling or shear mixing devices that cause the ingredients to collide and adhere to each other, producing larger particles with high-density. **Dry mixing** is used to blend dry raw materials to produce detergent powders and may sometimes involve the addition of small quantities of a liquid. The finished products from the above processes typically undergo further unit operations including storage in silos and bags, filling into bins and hoppers, handling and transportation in conveyors, packaging, etc. Key unit process challenges to be considered in the integrated fablab design will include:

Powder Mixing & Segregation: Many raw materials in detergent production are particulate form and their mixing/blending is a core unit operation. If the ingredients have different particle properties, in particular, particle size, this may cause segregation or de-mixing after mixing. This problem can seriously limit the potential for creating ingredient mixes. Online monitoring techniques (PAT, Chemometric data analysis, soft sensors, etc) will be developed and used for determining the endpoint of mixing processes. Producers would like to have tools for measuring segregation tendency and for predicting the maximum difference in particle properties tolerable before segregation becomes problematic.

Powder Caking, Scale formation and Fouling: During some unit operations (spray drying, conveying, mixing, etc.) with specific formulations, powder may stick onto and coat the wall of the equipment and conveying lines. In powder mixing, the particles may stick to the mixing chamber surface and form a crust over time. This causes major problems as the equipment need to be cleaned to avoid blockages, cross contaminations, etc. There is therefore a need to integrate testing that can predict the likelihood of powder sticking onto a surface, and how this is affected by the composition of the powder and processing

conditions. Modifications to the equipment surfaces and pipe walls (e.g. coating) could also be implemented to mitigate these problems.

Powder Storage: Storage of raw materials, intermediate ingredients and final products plays a vital role in production processes. Frequent blockages in silo hoppers, uneven discharging, silo flooding, etc, are some of known challenges. In addition, powder storage devices contribute to segregation issues in downstream processes. ***Powder Conveying:*** Conveying of powdered materials, as raw feeds, intermediate products and final products in packaging plays an important role in terms of product quality, energy consumptions, health and safety issues, etc. Conveying pipeline blockages, product contamination and spilling out due to erosive wear of pipelines, energy optimisation, etc are some of the major challenges couple with powder transport.

Liquid personal care products manufacturing fablab design considerations: The just-in-time manufacturing of liquid personal care products such as shampoos also bring a specific number of challenges that are highlighted below. The main operations that will be required for the formulation of liquid products are **liquid/liquid blending**, **solid/liquid incorporation** which comprises both dissolution of soluble species and dispersion of insoluble particles along with the related wetting and incorporation as the two phases come together. Related to these two major formulation steps are powder conveying and dosing (discussed extensively above), rapid heating and cooling of components and products, liquid storage and stability, powder pumping and dosing, and finally cleaning. The challenges associated with these are discussed below.

Blending of viscous liquids: Liquid/liquid blending of multi-component viscous systems can be difficult to achieve quickly with non-turbulent mixing often being required in order provide an optimal shear rate that does not disrupt the rheology of the formulated product. Moreover, the blending of multiple liquid components with different rheologies can be difficult to achieve in:

Powder/liquid incorporation and dispersion generation: Many FMCG will require the addition of solids to the liquid system. This can be for substances that dissolve or insoluble substances that need to be dispersed. In both cases, ensuring a favourable wetting and incorporation of the solid phase by the liquid will be key in ensuring rapid processing. From a process perspective, this operation will most likely be the most difficult to achieve with only a limited amount of optimisation being possible. To prevent unacceptable processing times for manufacturing within fablab, where possible, concentrated dispersions or feedstocks will be employed,

Liquid pumping & dosing: Efficient pumping of the feedstocks and dispensing of the product will be required by a liquid processing fablab, this will need to be conducted in such a manner as to not compromise the key quality attributes (rheology and functionality) of the product. While pumping of highly viscous liquids is relatively well understood, even at small scales, the design of the fablab will need to be sufficiently robust to prevent unwanted build-up or sedimentation within pipework.

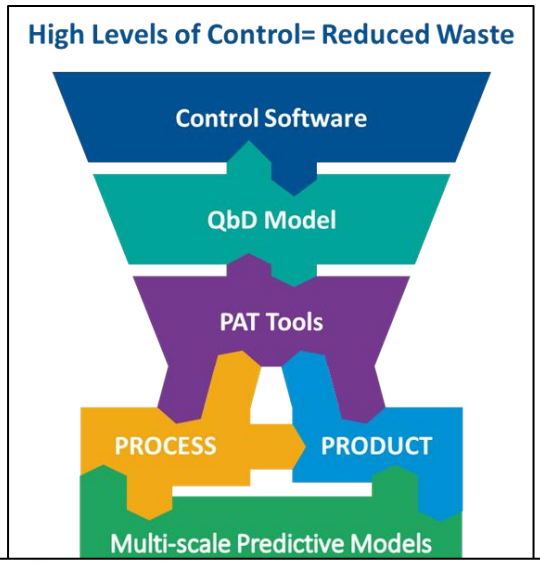
Liquid storage: Storage of multiple liquid components will need to be considered within the formulation of liquid products, along with vessel recharge, environmental control, and shelf-life. The challenges associated with achieving this will be significantly different from the strategies employed for powder handling be equal within ***Cleaning***: Cleaning of dispensing lines and mixing tanks will need to be conducted following the formulation of each unique product. Strategies will need to be developed in order to minimise the amount of waste or off-spec material generated by the fablab.

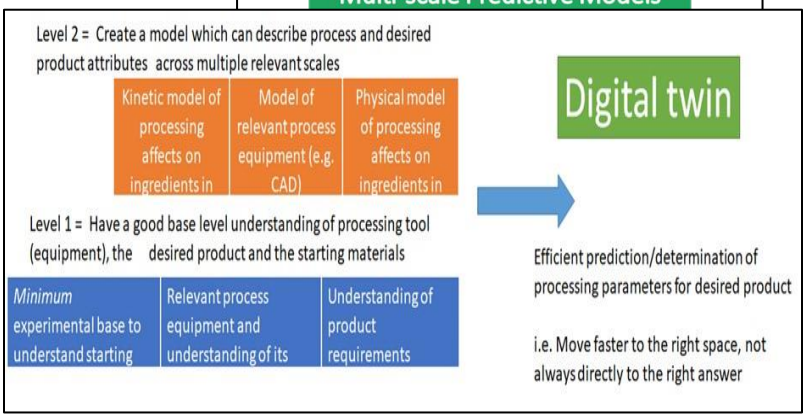
The above unit operations and key aspects of the powdered and liquid fablabs will be modelled and simulated using CFD by DSC, then the key unit operations validated by CPI and STEF prior to integration.

Fablab High speed analysis and quality control: Process analytical technology (PAT) is a well-established concept for providing information about products and processes. While such technologies are often not in themselves 'new' the extent to which they can be miniaturised to be accommodated for process monitoring or inline material characterisation and quality checks is a strong drive for their

continuing development to open up new areas of applications. It is believed that the fablab concept will pose significant opportunities within the field of PAT and hence drive the design and uptake of the next generation of in-line sensors that advance the state-of-theart. In addition to PAT being used for a 'quality check' of materials as they are being dispensed to the customer, it is envisaged that feedback from the sensors can be used to optimise the manufacturing process. While the fablab process is inherently batch driven in nature, by employing PAT analysis of the materials as they feed in to the process, coupled with analysing the product as it is dispensed to the consumer, over time, the neural networks that provide the self-learning capability of fablab will be able to predict (based on the raw material inputs) the optimum way to run the process at the heart of fablab device. Over time, this approach will provide significant benefits as the fablab continues to learn: firstly, it will ensure that the properties of the product being delivered to the consumer are optimal and will hence reduce the amount of off-spec material being generated, secondly, it will optimise the process such that the product can be delivered to the consumer in the shortest possible time, finally, it will make the fablab robust via building predictive process models such that it can maximise its robustness in being able to deal with variations in the quality/properties of feedstock materials while still generating a formulated material of quality (model predictive control). While such techniques are ubiquitous within continuous processing they can also be used for a batch-driven manufacturing by providing predictions on processing times and endpoints etc.

Digitisation of formulation manufacturing: In order to complement the PAT integration within fablab, by using the coupling of multi-scale models, a framework for building a 'digital twin' of each fablab will be generated. Broadly, a digital twin will employ multi-scale modelling in order to link well established predictive models to generate insight as to the optimum way for a process to be run. While the PAT is able to learn over time to ensure quality products through manufacturing excellence and process robustness, the digital twin will allow the fablab to predict how it should operate when it is required to run with feedstocks for which it has little or no data. For example, it is envisaged that the formulation space of a particular fablab will not remain static and will evolve over time as formulators look to incorporate new and innovative components within their products. Hence, by predicting how these new components will behave and interact with the established formulation space, the digital twin will ensure that fablab is incredibly robust in being able to maintain the generation of optimum products with little historic PAT data on a new component. Moreover, the digital twin will allow DIY4U platform formulators the ability to test how a new component could be incorporated within a fablab without the necessity for running a large matrix of compatibility experiments. By linking neural network driven learning, PAT for quality and predictive control, and a digital twin manufacturing approach, fablab will possess the required tools to drive the digitisation of the formulation of fully personalised consumer goods.





Inefficiency barriers of small batch/customised products: In addition to using the PAT and control approach to improve efficiency, the economic feasibility of the integrated fablab will be accessed by performing a detailed study around process efficiency (optimised production rates), inventory costs, changeover/downtime and other relevant factors.
Customisation and registration/consumer safety of a changing product: The consortium will use input from Product Stewardship teams from the relevant industries (e.g. P&G) and work with relevant regulatory bodies (e.g. AISE) to develop a software mechanism for evaluating the safety level of a requested product, developing MSDS, safety labelling etc. The DIY4U design algorithms will ensure that only inherently safe formulations can be designed. A request for a potentially hazardous or unsafe formulation would automatically return the message "NOT SAFE/POSSIBLE".
***High speed labelling (for both ingredient labels/customisation*):** High speed labelling will be required but this does not need to be high throughput. Labelling will be provided to enhance the consumer experience and also to provide full materials and process traceability for products produced by a fablab. It is understood that current state-of-the-art within labelling will be sufficient to provide this with the novelty existing in the application to fablabs.

Fablab Development: CDY (robotic and automation solution provider with expertise in manufacturing automated equipment for the formulation industry) will work with consortium partners (P&G, CPI, STEF and CTM) to manufacture fablabs meeting design specifications and incorporating early unit process testing and simulation activities performed at CPI and STEF. The fablab should provide simplified simulation tools to facilitate the manufacturing process to the end-user (e.g. the consumer). CTM will implement a simulation module tools aiming to drive the user towards an easy product manufacturing. DSC will additionally implement DIY4U platform CFD animations of the internal performance of key operational units that have been virtually twinned.

Fablab Demonstration, Optimisation and Training: Following the installation of the developed fablabs at CPI and STEF, both partners will act as MDFs for the fablabs. The MDFs will provide open access for stakeholders that have designed new customised or personalised products on the DIY4U platform to use the fablabs to manufacture and characterise their products. Feedback from users will be used to optimise both the fablab and design & modelling tools on the digital platform.
The MDFs will provide training to fablab users as well as to students and other industries interested in the DIY4U approach. Dedicated DIY4U training programmes/events (student placements, apprenticeships, workshops, conferences, equipment training and use of digital tools like virtual and augmented reality) will be organised and delivered to prepare current and future EU workforce for embracing the revolutionary customisation and smallscale industries approach.

Open access element of the MDFs: During this project, the MDFs will provide open access to all EU FMCG stakeholders (including consumers and SMEs) that have successfully collaborated on the DIY4U platform to create a new customised/personalised product design that they would like to manufacture using the fablab MDFs. From M24, DIY4U will open access to the public and FMCG value chain through a public event and via digital marketing strategies to invite stakeholders to use the DIY4U platform to design new customised/personalised detergents and soaps. In parallel, PICOM will organize a co-design workshops with 15/20 people including supermarkets, FMCGs and selected customers 1) to assess process and solution of both Open Innovation B2B/B2C digital platform and MDFs and 2) to detect potential improvements. From M24-M36, optimal formulations designed and digitally tested on the platform will undergo a selection process that will choose customised/personalised formulations that will be manufactured and tested at the open access MDFs for free. The open access MDFs will be used to validate the DIY4U concept (i.e. going from product idea to prototype formulated product within a few days) and the overall DIY4U sustainability (demonstrating the potential cumulative profit of >9M € within 7 years post project (section 2). The detailed procedure for selection process is summarised

below. A DIY4U OI Selection Committee (OISC) will ultimately be responsible for selecting the formulations that will be manufactured for free. The DIY4U consortium has allocated adequate resource and budget for STEF and CPI budget to provide free open access to third parties selected via OICs to develop and manufacture new customised DIY4U formulations. This free open access budget will cover the cost of training the third parties and providing DIY4U manufacturing support and consumables. The selection criteria will ensure that the third parties provided with free open access to the MDFs comply with EU rules. Third parties that have designed viable optimised formulations but do not meet the selection criteria for free open access to MDFs could self-finance their use of the open access MDFs.

How we will select formulation recipes to be manufactured at the MDFs: Companies applying to manufacture their formulations at MDFs will provide 1) company description, 2) marketing approach (finished product features and differentiation from customer's point of view, market analysis, targeted business canvas) and 3) tests description (procedure and involved stakeholders during testing, overall planning, external (i.e. CPI and STEF) expertise, resources and budget needed during MDFs use, internal dedicated resources, key indicators to be measured during tests, action plan in case of success). The consortium and its Advisory Board will propose eligibility of applicants based on company profile and proposal completeness. The DIY4U Selection Committee will select formulations and companies according to three criteria: product innovation, business model and ability to address market needs, ability to perform testing.

Following the completion of the DIY4U project, the MDFs will be available for EU companies that wish to use it on a flexible 'pay-as-you-use' collaborative/private basis, to manufacture customised/personalised FMCG. Companies needing to use the MDFs will be provided training that allows them to operate the fablab themselves. Fablab MDF users will be responsible for the use of products (e.g. for relevant studies) manufactured. The facility will be maintained to standards, including appropriate cleaning protocols, and record keeping to allow flexibility for varying users as well as varying materials that feed through the infrastructure.
To further encourage the use of the facility by multiple organisations, confidentiality issues between users will be addressed by following best practices established within the respective operating open access innovation infrastructures at CPI's National Innovation Centres and STEF. Both organisations will implement their respective policies for conflict management, confidentiality and IP, allowing multiple users to use the facility without fear of losing their IP or staff being conflicted. Appropriate resource planning (based on conflict checks and expertise) and effective project/innovation management by experienced project and innovation/technology commercialisation teams at CPI (>600 years combined experience) and STEF (>400 years combined experience) will also ensure adequate handling and management of partners' confidential information at MDFs.

Business needs and use cases definition and assessment … and be agile to react on evolutions: Business partners interviews within the consortium, studies analysis, external brands and retailers' interviews will be used by the project to find out business partners and ecosystem needs. Co-creation process will be used to design use cases and identify relevant initiatives that should be considered for collaboration. This process will work on some specific issues including: Which stakeholders are about to emerge? Which stakeholder should operate the platform and the fablab (manufacturer, retailer, city, third party, …)? i.e. what is the right value chain? Which stakeholder is in charge of customer relationship? Is there a need to coach the customer during the configuration process (because choices possibly impact health, price, environment, …)? Key performance indicators will be defined in 3 domains (business economic impact, customer satisfaction, process improvement) and assessed during pilots through measures and customers feedback. The use cases descriptions will be used 1) to specify IS architecture; 2) to adapt project's activities as well as 3) to adopt solutions adopted by the community.

This approach will support improvement and enhancement of project outputs in iterations. The approach intention is to ensure that the final outcomes are well aligned with the latest trends and achievements.

Use open innovation competitions (OICs) to extend DIY4U functionality, test and promote DIY4U: To promote SME participation and unlock new business opportunities for SMEs based on the DIY4U approach, we plan to run two Open Innovation Competitions (OIC), one launched around M18 and the second around M28, to enable gradual and iterative expansion of DIY4U platform, and to promote the use of the DIY4U to create new business opportunities for SMEs. A standard process has been defined to select up to 10 Open Innovation Projects. The process starts with defining a brief for the OIC and promoting it in social media, newsletter, events, and meetups with partner organisations in the relevant communities. After receiving a sufficient amount of applications, a curation process starts with follow up calls to select the best SMEs. Finally, this cohort is invited to a 1 day event with guest speakers from the consortium and workshop-like activities to start defining ways to collaborate, augment the DIY4U platform and develop bespoke formulations for commercialisation. This initiative will be useful to:

1) Get feedback and test the functionality;
2) Get businesses that can become contributors to data;
3) Use the platform and fablab MDFs services to create and make new/innovative bespoke FMCG that they which to commercialise;
4) Improve the solution by adding new technical modules (if partners consider there are significant technical improvement areas);
5) Evaluate new business models, for the engineering of personalised goods, particularly for SMEs that can adapt to rapid demand changes and can implement changes in a shorter time to market; 6) Identification of new use cases for the personalisation or fabrication of other goods;
7) Reduction of design and development costs due to co-design and co-development between fablabs; 8) Promote the resulting solution towards B2B clients.

We expect to fund up to 16 SMEs by providing real cost of up to 50k€ per successful application in addition to relevant technical and business mentorship from within the consortium to help the winners undertake the proposed OIPs. M15 OIC (200k€, 4 funded OIPs) will target SMEs that have developed relevant digital solutions that can be integrated and tested with the DIY4U platform. M22 OIC (300k€, 12 funded OIPs) will target (minimum Eligibility Criteria) EU SMEs aiming to: (a) use the platform and fablab MDFs services to create and make new/innovative bespoke FMCG that they which to commercialise; and (b) explore the use of DIY4U services for a new business offering (e.g. small-scale manufacturer) in the FMCG sector. The OIC selection process (further details in section 3, WP9) will follow: RFP and answer form definition (by DCC and PICOM, All partners and Advisory Board), communication and interaction with relevant ecosystems (by All, led by DCC), proposals acknowledgement and eligibility instruction (by DCC and PICOM), proposals evaluation upon business and technical aspects (by 3 external evaluators), selection of proposals (by DIY4U Selection Committee), notification to winners (by DCC). Each selected project will be supported by relevant DIY4U partners in order to ease use of DIY4U resources, follow-up steps and planning, review deliverables and results and monitor overall. An overall evaluation of OIPs vs DIY4U solution and ecosystem will occur at M36. **1.3.3 Technology Readiness Level**: The start and end TRLs of DIY4U

Technology	Start TRL	End TRL	Expected advancement through the project

Knowledge Management and Collaboration Platform	4	6	The KMC will use as foundation the platform developed by IRIS in the project IBD (see relevant project below). In the context of DIY4U IRIS will adapt this platform using the cloud architecture employed during project SHAREBOX for a cloud online deployment. Both IBD and SHAREBOX platforms have been developed to TRL7 but the adaptations necessary for DIY4U will bring TRL down to 4. However, by the end of the project, the demonstration of the KMC with end-users will validate it at TRL6
Knowledge-based formulation design	4	6	Elaboration of Knowledge base through real user experience and digital formulations in order to propose safe and trust formulations, combining background knowledge and IP using established technology to demonstrate a new system to TRL6.
Formulation and Process Modelling	4	6	Formulation and process modelling are well established in industry. DIY4U will provide a step further so non-specialist/DIYers/SMEs will be able to provide better and safer products to consumers.
Advanced data analytics	4	6	Enhance the collection of data and the extraction of valuable information from different types of data. Also, improve availability and scalability of services.
DLT/Blockchain	4	6	The principles of IP tracking, uniqueness of digital data, and request processing are already validated/established (TR6-9). Since these have not been integrated and validated with the DIY4U/KMC platform and related components, relevant adaptations are required, hence it appropriate to consider that DIY4U DLT/blockchain platform will start at TRL4 and reach TRL6 at the end of the project. DIY4U will develop API integration to ERPs, standard data formats for production requests and interactions with 'offchain' data storage systems, as well as governance processes between parties sharing the system. If regulations and banking behaviours change suitably during the DIY4U project, we will also enable a blockchain-based payments layer.
Fablab	4	6	Most of the hardware and electrical components for the the fablab are already at TRL6-9 and CDY has built several commercial and prototype systems for other applications using similar components. DIY4U will be the first miniaturisation and integration into a fablab prototype machine (using background IP from CODY, STEF, CPI and P&G) followed by its demonstration (i.e. achieve TRL6).
Fablab PAT and automation	4	6	VTT has developed relevant PAT tools to TRL5-7 (e.g. NIR spectrometers). DIY4U will develop the integration of the PAT tools into the fablab and study different sampling interfaces, as well as data analysis methods to reach TRL 6 for the integrated system at the end of the project.

1.3.4 Trans-disciplinary considerations: DIY4U's collaborative approach will provide all its partners access to the interdisciplinary knowledge required to develop the digital platform and fablab. The project has a transdisciplinary character as it will incorporate software engineering, ICT, material science, physics, chemistry, mechatronics, chemical engineering, process technology, numerical simulations, environmental science, marketing and finance.

1.3.5 Other national and international research and innovation activities linked with DIY4U

TagItSmart! - A Smart Tags driven service platform for enabling ecosystems of connected objects (https://www.tagitsmart.eu/) **(PICOM)** – The project includes 3 types of end-users: brands/ manufacturers (Unilever), retailers (Leroy Merlin, Univerexport) and service providers (Dondelotiro).

PICOM is involved in use case definition, tests follow-up, showroom creation, dissemination workshops, exhibitions, standardization and open call management.

Commercial projects (PICOM): Delivered several confidential product customization projects with manufacturers and retailers including chocolate 3D printing, color-changing shoes, customized 3D printed heels. **Synchronicity (Digital Catapult) - H2020** - Converging EU IOT standards and creating a blockchain-based panEU marketplace for IOT data exchange, in pilot stage with 8 cities.

Content Personalisation Networks (Digital Catapult) - H2020 - Creating a GDPR-protecting personal news reader app complemented by a blockchain-based pan-EU pool of licensed media content to lower barriers for freelancers to connect with major news agencies and vice versa.

DLT Field Labs (Digital Catapult) - Virtual collaborations across the UK to develop and evaluate blockchainbased systems beyond proof-of-concept stage. Each project involves a number of industry partners combined with DLT-developing SMEs/scale-ups to effect knowledge transfer and publicly demonstrate the value proposition of this technology.

EFF has contributed to several architecture and development commercial projects covering adaptive state sharding **(Elrond)**, enterprise blockchain **(Insolar)**, blockchain drone management **(SkyyNetwork)**, blockchain business connectivity **(Maxonrow),** digital identity **(Helix).**

IbD -Intensified by Design platform for the intensification of process involving solids handling (IRIS, Analisis-DSC, STEF, VTT): H2020-IA, SPIRE-08-2015 – **Online platform for Process Intensification (PI) of chemical process** so that among others advantages of batch processes are transformed into continuous moreefficient process.

SAM - Self-Adapting Model-based system for Process Autonomy (STEF) - The project is granted under **BIAIPN fund program** of Research Council of Norway (**RCN**) with 8 industrial partners.

BIA- Effective Powder Handling (STEF) - BIA-KMB-RCN Minimising scale formation and erosion in pneumatic conveying pipeline of bulk powders

MFLOW- Non-intrusive mass flow measurement & control system in pneumatic transport (STEF)- FORNY-3-RCN- commercialising of non-intrusive mass flow measurement system for pneumatic conveying of particulate bulk materials.

BIO-Powders (STEF) - Marie Curie Research Training Network (Project ID: 512247) **EU- FP6**

AceForm4.0 – Value Chain in Formulation Manufacturing (CPI): H2020-CSA (Contract No. 723045) – Facilitating exchange of non-competitive know-how in formulation technologies between stakeholders operating in key sectors that are dependent on formulation innovations and associated automation and digital tools.

ADDoPT – advanced digital design transforming pharmaceutical development and manufacture (CPI): InnovateUK – Developing and implementing advanced digital design techniques that eliminate nonviable drug candidate formulations, streamlining design, development and manufacturing processes.

PROSPECT-CL (CPI): Innovate UK – National Formulation Centre Strategic project in collaboration with UK industry – Development of complex liquid mixing scale up rig with advanced process analytics and advanced process control.

PROSPECT-CP (CPI): Innovate UK – National Formulation Centre Strategic project in collaboration with UK industry – Implementation of PAT and Advanced Process Control on Twin Screw Wet Granulator for pharmaceutical and non-pharmaceutical applications

PROSPECT-P&F (CPI) Innovate UK – National Formulation Centre Strategic project in collaboration with UK industry – Implementation of flexible rig for studying powder flow and packing

ModelsMPP (CPI): Innovate UK – National Formulation Centre Strategic project in collaboration with UK industry – Transferring innovations in particle modelling into industrial practice - Methodology and framework to accelerate innovation in formulation development and manufacture cross-sector

Development of AI/data-mining and web-based applications (Python, R) (RDI): Veolia, EDF, Fareva

CFA. Advanced Manufacturing Centre for Aeronautics and Naval Sectors (CTM): Innovative SPANISH – Implementing advanced manufacturing techniques (3D printers, fablabs, robotics, design and simulation, etc) for Aeronautics and Naval design and manufacturing components.
DIH FAT. **(CTM)** Digital Innovation Hub (DIH) Advanced Manufacturing in Transport Sector. CTM is the coordinator of the Andalusia Regional DIH strategy. This project is focused on establishing a platform where manufacturing innovations can be developed by SMEs.
SHiELD (STL): H2020-RIA, DS-03-2016 – Research and development of data protection and privacy framework for security toolkit for data exchange systems like OpenNCP
OPERANDO (STL): H2020-IA, DS-01-2014 – Data protection and privacy platform to enable users to control the use of their data and to enable the marketing of privacy service provision
eWALL (STL): FP7-RIA, ICT-2013.5.1 – Research on ethics, security and privacy by design in sensing and cloud systems for personalised healthcare services
Safe Life Design of a Mixing Vessel for detergent production Persan (Analisis-DSC): Henkel – Safe life design of a mixing vessel for detergent (non-newtonian material) production.
Thermal and fluid-dynamics analysis of a tensoactive agent production process (Analisis-DSC): This study evaluates using CFD tools, the behavior of three in-line devices for a tensoactive production process.
Optimization of a reactor vessel and a heat exchanger for detergent KAO Corporation (Analisis-DSC): **KAO Corp** –The project aims to optimize the production of surfactant batch reactor analysis to avoid residual reaction in stirred reactor batch type using fluid dynamics optimization (CFD), and making proposals for improving efficiency.
ProPat http://pro-pat.eu/ (VTT, IRIS):– Online PAT platform - aims to develop novel and smart sensors and analysers for providing measurements on composition, particle size, local bulk properties and process parameters. The platform also provides self-learning and predictive capabilities aimed for dramatically reducing over costs derived from even slight deviations from the optimum process.
CODY Commercial Project: Inspection and Packaging Line for radiopharmaceuticals. Recipe handling, data logging, electronic signatures and reporting with TITAAN software.
CODY Commercial Project: Product test station for testing high-voltage equipment. Automatic test execution by retrieving test recipes from external database using software developed by Cody (TITAAN). **P&G UK led a Chariot UK AMSCI £18Million project, powder processing grant**
P&G UK was involved in the FP7 Marie Curie ITN project PARDEM & TMAPP, hosting 1 ESR.

1.3.6 Sex and/or gender issues: DIY4U consortium partners recognise and wholly support the EC's policies/priorities for Promoting Gender Equality in Research and Innovation (R&I): women participation in research must be encouraged to close the gender balance gaps in research teams; promoting gender balance in decision-making to achieve 40% of the underrepresented sex in panels/groups and 50% in advisory groups; and integrating gender dimension in R&I content to improve scientific quality and societal relevance of generated knowledge, technology and/or innovation. The general under-representation of women in FMCG sector R&I makes Gender issues particularly important for this project.

All DIY4U participants are equal opportunity employers, with non-discriminatory policies with regards to gender. Some DIY4U participants (e.g. CPI with 26.66%) have higher proportions of female science and engineering staff than the national or regional (EU) industry norm. In accordance with the above EC's recommendations for Promoting Gender Equality in R&I, we have actively included women in the technical delivery of the project's WPs and tasks and expect to reach ≥25% of our total expected number of project personnel. All WPs are genderneutral with respect to employability. DIY4U digital platform and fablabs will also be gender-neutral, allowing anyone to become a creator/maker of particulate/liquid FMCG irrespective of their gender.

1.4 Ambition

DIY4U will bring to bear a truly "Disruptive Business Model" that the FMCG companies fear and want at the same time. It requires digital developments that ultimately enable materials supply to be more distributed than today and in a more open innovation format that allows consumers to design and customise their own products. This will deliver an unprecedented scale of "designed for me" and "Sustainability" in the FMCG sector, with less waste/transport/CO_2 emission and improved efficiency. DIY4U digital platform combined with its fablab services will be the first integrated digital infrastructure available for non-FMCP experts or consumers to design and manufacture their own particulate/liquid FMCG based on their individual preferences. The nearest current stateof-the-art to our DIY4U approach are Function of Beauty that has built a software platform to determine precise formulas for both shampoo and conditioner for every customer and Proven that uses data from the world's largest beauty database and machine learning algorithms to personalize skincare products. Also, paint customization services are offered by manufacturers (e.g. AkzoNobel), supply chain stakeholders (e.g. Riolett Custom Aerosols) and retailers (e.g. B&Q, Halfords). DIY4U surpasses the paint and the beauty customization approaches for the following reasons:

- Paint customisation services are only really to customise colour and it is not possible to request other features such as a custom sheen (gloss) level. Moreover, paint customization services only apply to a limited product range (essentially broad wall emulsions). DIY4U will provide unrivalled customization/personalization for a wider range of soft matter FMCG.
- DIY4U provides open source tools (via its digital platform) that allow anyone without FMCG formulation expertise to digitally design and test the performance of personalised/customized formulations. Paint and beauty formulation knowledge is currently locked in proprietary manufacturers' formulation modelling capabilities or databases that are not necessarily suited for use by non-experts or consumers. DIY4U will provide a user-friendly digital knowledge management open access resource that displays relevant formulation ingredient information and ensures compatibility of modelling and simulation with different process information systems, enabling anyone to formulate new safe and environmentally benign products.
- The DIY4U platform capability that allows manufacturers to plug-in their models (e.g. soft matter, process, LCA/LCCA) for consumers, DIYers and SMEs to use for creating bespoke formulations is a key differentiator of DIY4U. Formulation computational modelling including a series of numerical methods will forecast how the formulation will perform in the consumer home, reducing the testing procedures and avoiding a bad experience to the consumer. Formulation process digital twin will virtually create products, mimicking the real equipment performance. The digital twin will take advantage of CFD tools to digitally analyse and optimize the formulation processes.
- DIY4U's big data analytics/AI capability would allow for real-time tracking and innovation respond to trends in consumer habits and associated stress/strain on upstream supply chain.
- Fablabs will be designed such that it can be operated in home, in shop or in small factories by anyone, unlike specialist machines used for paint customisation by trained technicians. Furthermore, the connectivity between DIY4U digital platform and fablabs will ensure real-time analysis of consumers' demands and facilitate rapid production of particulate/liquid FMCG tailored to meet changing consumers' needs. This will create business opportunities for new actors, especially SMEs, in the FMCG sector.

Data Analytics Innovation Character: The main innovation of the data analytics module is the strong consideration of the human factors such as perception, culture, ontology, semantics and cognitive aspects. The DIY4U project will take into account all these complex parameters to define a "systemic

design"[8]. Moreover, a smart mix of algorithms and methods will be proposed while keeping the application fast and effective. The developed human-centred application will understand the needs, have the capacity to analyse and extract valuable information from any kind of data (image, text, video, process, thinking and audio …) and support the change of demands in order to ensure sustainability and scalability of the DIY4U project.

Modelling and Simulation Innovation Character: DIY4U will deliver innovation of DIY design by implementing the predictive simulation techniques (feed forward) based on multivariate analysis to predict flow, structural and thermal features, and the behaviour of ingredients in the formulation process at each process step. Currently, process systems in formulation fail to generate real-time data sets that enable predictive design, i.e current process optimisation models are not fit for on-demand customization (feedback optimisation only). DIY4U will use Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) to provide structural, thermal and fluid analysis of the formulation supported by a digital reproduction of the physical process to optimise cost and environmental performance, taking customer preferences and behaviour change into account. The fablab digital twin (a key innovation feature of DIY4U) will make use of the fablab prototype equipment to ensure the virtual twin is correctly validated with the actual equipment. The developed DIY4U open source LCA/LCCA modelling capability could be used with any soft matter FMCG to help consumers and SMEs make informed

choice based on environmental performance. This is a new innovation to the market that is hoped to positively influence human behaviour before committing to the manufacture of goods, by increasing awareness of a product's sustainability performance, which may drive a change in consumer purchasing habits.

Blockchain Innovation Character: **DIY4U will be the first publicly demonstrable application of blockchain within the formulation/FMCG manufacturing industry and value chain.** According to Venture capitalist William Mougayar, blockchain is "the second significant overlay on the internet, just as the web was the first layer back in 1990". Within the DIY4U platform, blockchain will enable managed access to valuable assets, such as intellectual property, new FMCG formulation innovations and payments, without the need for intermediaries such as IP Attorneys/Lawyers. Blockchain/DLT provides an irreversible, secure and time-stamped record, enabling the DIY4U platform to be used for managing access to data and IP provided by multiple parties in a granular fashion. It can also provide evidence of use of company or individual IP (e.g. formulations, algorithms) and enforce IP licences through smart contracts.

DIY4U will also pay close attention to the financial regulations emerging in the EU and G20 around the peer-topeer transfer of value through blockchain. Whilst information currently travels instantly around the globe, the same cannot be said about business transactions and exchange of value. A typical international payment takes 3-

5 days to settle, with ≥5% error rate and average cost of €37; €158.34 trillion worth of international payments are made annually at a combined cost of over €1.5 trillion p.a. If EU regulations change during the project to allow this functionality, and banks permit API integration with their services, payments for DIY4U could be automatically facilitated by 'smart contracts'[9] - code executed on a blockchain - and completed instantly, allowing FMCG value to move globally just like information.

[8] Jones, P. "Systemic design principles for complex social systems" volume 1 of the Translational Systems Science Series, pp 91-128, 2014, Springer Japan.

[9] https://ripple.com/insights/the-internet-of-value-what-it-means-and-how-it-benefits-everyone/

Fablab Innovation Character: The fablab is in essence an automated liquid or powder mixing and dispensing system that demonstrates self-cleaning and quality assurance. The novelty of such an approach will be embedded in advancing the limits of miniaturisation in terms of feeding, blending, dosing, and process analytics. While inline product testing using process analytical technology is itself not a new concept, achieving it within an interlinked complex formulation network around a geometrically constrained manufacturing space, combined with a limited analysis time will require innovative approaches. The integration of real-time process monitoring, complemented by decision making algorithms and self-learning neural networks will allow the fablab to provide unrivalled 'right first time' small-scale manufacturing machines for soft matter FMCG.

Patent/IP Search: A patent and IP landscape review performed by IP experts at STEF and STL suggest that DIY4U digital platform and fablab are novel and patentable; and will have no freedom to operate issues. The closest recent patents identified are:

- Customized fast-moving consumer goods production system (WO2017014696A1): is the nearest patent to the DIY4U solution but it does not propose a collaborative platform or smart algorithms for formulas.
- Custom cosmetic blending machine (US8564778B1): is only devoted to specific cosmetic products and cannot be deployed for wider FMCG or other sectors. It also requires operation by experts.
- Additive manufacturing for beauty products (WO2015132734A1): is an innovative system proposed by L'Oréal, however it is dedicated only for 3D printed nails.

This IP analysis proves that there is no system that proposes a large collaborative solution for DIY design and production based on open technologies.

IP held by the consortium: Digital Catapult holds IP on methodologies for reducing smart contract computation costs on the public Ethereum blockchain. IRIS holds IP on the IBD® and SHAREBOX platforms that will be adapted and used for DIY4U. EFF holds IP for methods on decentralized smart agents. RDI hold IP for connected clothes and patent pending for connected medical device. DSC brings extensive knowhow on material and process modelling and simulation. CDY holds IP and knowhow on formulation manufacturing and related machines and related software. VTT holds IP on relevant miniaturised PAT spectrometers (e.g. NIR). P&G holds IP on detergent and soap/other FMGC manufacturing processes, machines and related modelling and simulation packages. STEF and CPI bring IP and knowhow on operation of small-scale automated equipment for diverse formulation applications. CETEMET hold patents for device and method for laying cables, device and method for processing electrical cables, and knot-tying head.

2. Impact

DIY4U is a highly impact-driven project, with potential to revolutionise the European FMCG sector and other industries. The project will accelerate and de-risk the development and adoption of open innovation and fablab approaches for the collaborative design and manufacture of customised or personalised products. DIY4U will directly contribute to the expected impacts as set out in the Horizon 2020 Work Programme 2018 – 2020 under the work topic DT-FOF-05-205: Open Innovation for collaborative production engineering (IA). DIY4U will ultimately contribute in enhancing European innovation capacity and competitiveness, creating new market opportunities for EU companies and catalysing their growth, leading to socially important impacts (jobs creation and retention). DIY4U will deliver further social and environmental benefits elaborated in this section.

2.1 Expected impacts

2.1.1 Expected impact as set out in the work plan, under DT-FOF-05

DIY4U will contribute to the expected impact of DT-FOF-05-2019 as detailed in this subsection.

Establish Open-Innovation networks for manufacturing that support customer-driven production all around Europe: DIY4U Open Innovation B2B/B2C digital platform will connect consumers with the European value chain for particulate/liquid FMCG, providing them with digital tools and resources that allow all actors in the innovation ecosystem (including consumers) to participate in the co-design and co-development of new products addressing real-time consumer needs. This European knowledge management and design platform will facilitate the sharing of knowledge (e.g. ingredient information including regulatory compliance and safety data, process information) and enhance the ability of non-formulation experts (e.g. consumers) to digitally design and simulate the performance of new personalised or customized particulate/liquid FMCG. The platform's open source product data exchange and standard representations of products and processes, coupled with formulation modelling and simulation tools, will enable a 'plug and produce' design approach, allowing the European supply chain to collaboratively and rapidly respond to customer needs. Real-time analysis of feedback and demand from consumers (using integrated AI/data analytics tools) and sharing of such data on the highly visible and transparent platform will enable the European FMCG innovation ecosystem to efficiently collaborate on the design and development FMCG addressing the real-time customer needs/demand. The developed MDF will provide open access DIY4U fablab capabilities to European FMCG stakeholders (including SMEs, new actors and consumers), allowing them to manufacture the personalised/customized FMCG designed on the DIY4U platform. This will de-risk small-scale manufacture of personalised/customised FMCG, promoting the wider adoption of the DIY4U Open Innovation and customer-driven production approach across the European FMCG sector.

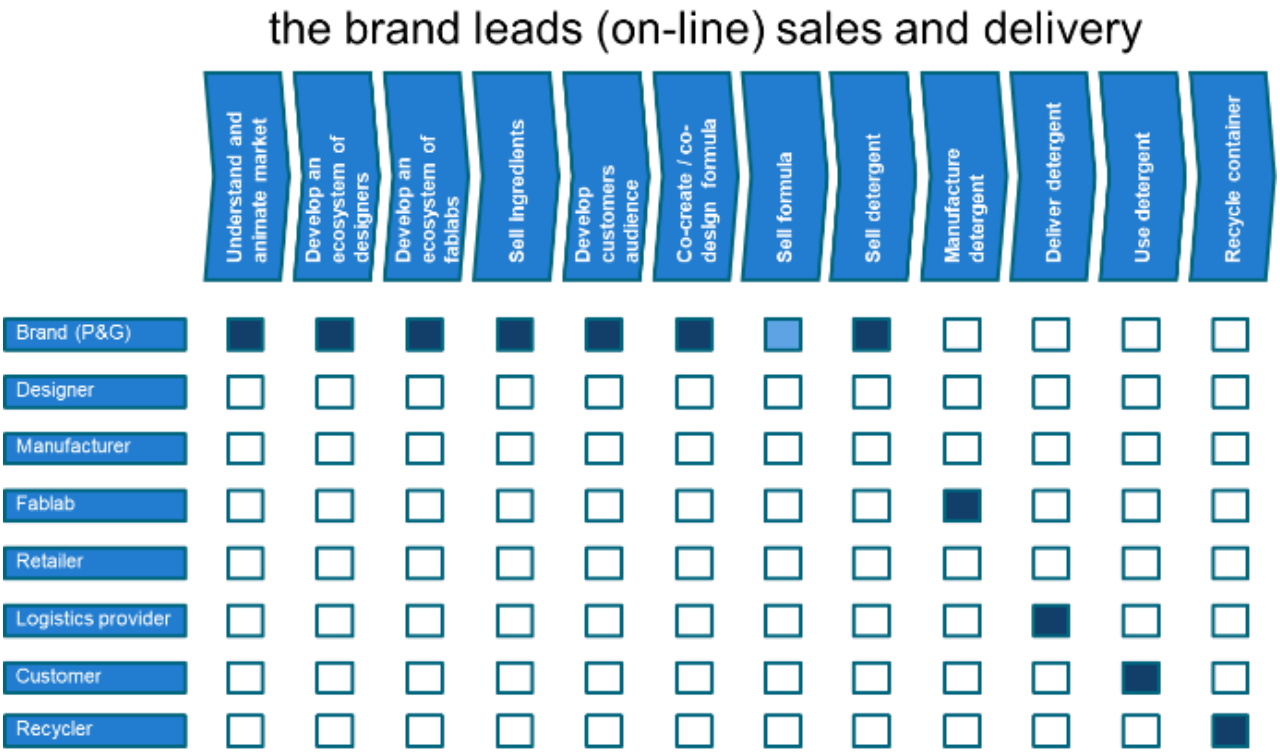
Creation of specific business models for the engineering of customised solutions, particularly for SMEs, rapid demand changes and shorter time to market: The project will develop business models that allow the European FMCG sector to adopt the DIY4U platform and fablabs for on-demand co-design and co-manufacture of personalised/customised products. The total perceived value of customisation can be described by:

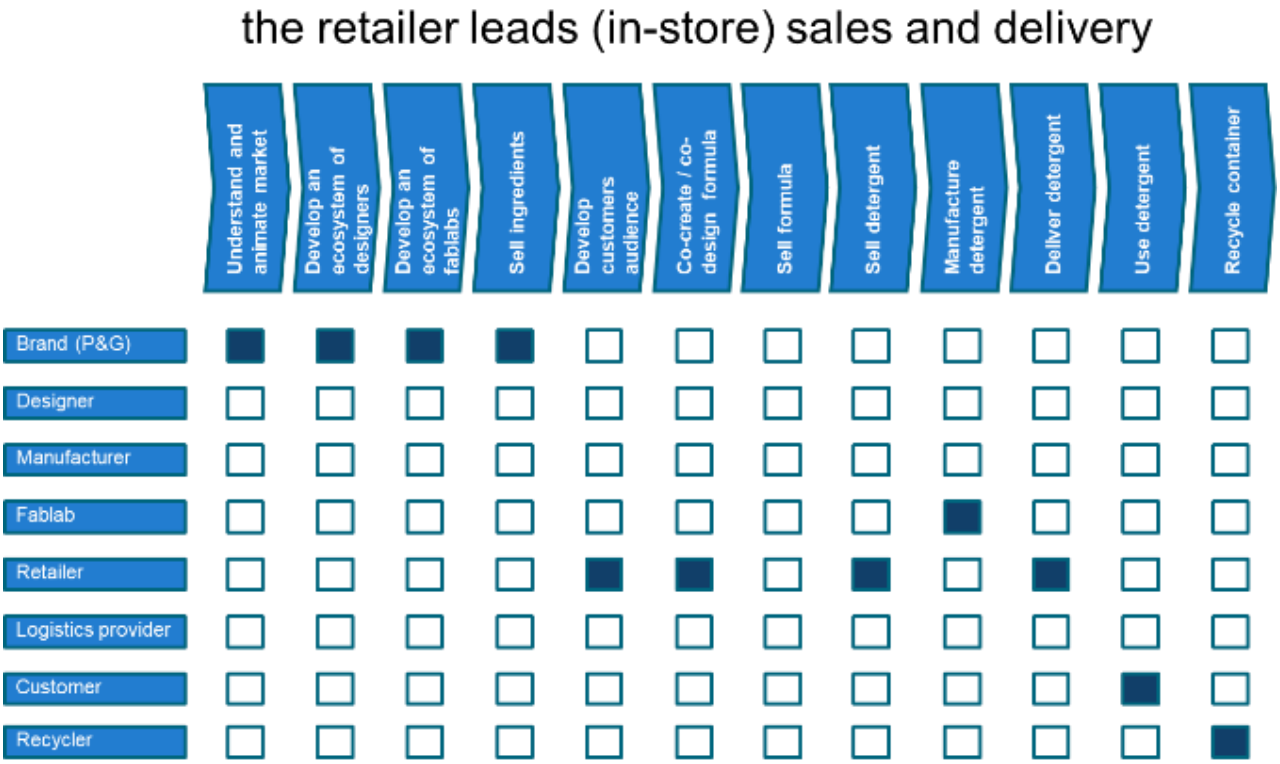
(1) Mass-customized product value, which is reflected in three dimensions: utilitarian, uniqueness and selfexpressiveness value, and

(2) Experiential co-design value, which is reflected in two dimensions: hedonic and creative achievement value. Studies show customers are willing to pay more for customised products: +30% for t-shirts, +200% for watches.

Within the business value chain, we consider 12 main activities (Understand and animate market, Develop ecosystem of (formulation) designers, Develop ecosystem of fablabs, Sell ingredients, Develop customers audience, Co-create/co-design formula, Sell formulation, Manufacture formulation, Deliver formulation, Use formulation, Recycle container) that drive to two realistic business models. In the first one, the brand leads (online) sales and delivery while in the second the retailer leads (in-store) sales and delivery for specific customers and/or micro-segments of customers.

Some specific stakeholders may be considered like
1) Designer, for formulation configuration and selling;
2) Service provider for using formulation (e.g. detergents for clothes washing or cosmetics for providing make-up services);
3) Recycler for reuse or recycling of out-of-spec formulations and containers.





Improvement of the co-design and co-development capabilities towards a reduction of development costs of new products and services: The DIY4U platform and fablab MDFs will facilitate co-design and

co-development of particulate/liquid FMCG, reducing development costs and time to market for new personalised/customised FMCG and associated services. The block-chain enabled digital platform streamlines co-design and codevelopment of new products whilst fablabs together with the platform make small-scale manufacture of personalised/customized products economically viable.

Increase of product variety and personalisation for higher customer satisfaction and loyalty: The DIY4U digital platform will be furnished with user-friendly capabilities and information that allows anyone to design and simulate the performance of particulate/liquid FMCG meeting their individual needs. This will facilitate product personalisation and increase product variety due to the participation of new actors in product design. DIY4U fablabs will make small-scale/on-demand manufacture of personalised/customised products practical and costefficient. Studies (Merle 2008) show that sales of customised products depend both on product perceived value and customer experience perceived value. Thus, the targeted co-design experience will provide high experience.

Improving the value proposition and governance models for DLT-based systems as part of the European Blockchain Services Infrastructure: DLT has been a topic of great interest across the EU due to its promise of creating new trust networks between sectors of the economy, as shown by the signing of the European Blockchain Partnership agreement by 27 EU national leaders in April 2018. DIY4U's DLT-based solution is part of improving the European Blockchain Services Infrastructure. The platform is intended to be open, permissive for other parties to extend and innovate upon, will enable secure data exchange related to FMCG, and increase the overall knowledge base of DLT across the EU. It will be one of the few at-scale deployments of DLT by the end of the project, proving the value proposition of this technology and therefore improving the business opportunities for hundreds of DLT SMEs across the EU who currently have to 'hard sell' to every new customer.

2.1.2 Improving innovation capacity, integration of new knowledge and EU competitiveness

DIY4U will enhance the innovation capacity of the European FMCG sector by providing it with the digital infrastructures and transformative business models required to pioneer the shift from large-scale manufacturing to industrial companies of DIY, fablabs and makers. A successful DIY4U project will give the European industry the confidence to further develop and commercialise the Open Innovation and fablab/DIY approaches after the completion of the project. Technical skills relating to the new disruptive design/manufacturing technologies and new approach to manufacture and supply will be transferable (within the consortium and externally). The DIY4U platform and open access fablab MDFs will serve as fertile grounds for strengthening the EU's IP base in particulate/liquid FMCG development and manufacture, through further development of background IP from partners and the generation of foreground IP within the platform and MDFs. Wider European adoption of the transformative DIY4U approaches will enhance the competitiveness of the European FMCG sector. Real-time analysis of consumer feedback and demand via the DIY4U platform will enable the European sector to respond in real-time to evolving customer demand and needs ahead of competing nations and regions (e.g. USA and Asia). The DIY4U digital platform will allow the generation and use of new knowledge and formulation design ideas coming from different and even new European stakeholders. This will lead to the creation of a wide variety of new innovative products for European and global markets. The new MDFs will offer fablab training and provide open access facilities for European FMCG stakeholders (especially SMEs) to manufacture and test new personalised or customised products. The digital platform and fablab MDFs will de-risk new product design and development and reduce time to market, hence enhancing the competitiveness of European stakeholders. Increased product variety and personalisation for higher customer satisfaction and loyalty (achieved thanks to DIY4U), will further enhance the competitiveness of the European sector.

DIY4U open access model and collaborative platform will allow for cross-fertilisation and exchange of knowledge in a pre-competitive high-risk innovation space. Its dedicated training programmes/events (student placements, apprenticeships, workshops, conferences, and equipment training) and use of DIY4U platform digital tools will help upskill the current and future EU workforce for academia, regulatory bodies and industry, further enhancing EU competitiveness.

The generic nature of technologies (e.g. multi-party process engineering DLT, plug in features, interoperability of modules and scalability of the KB) integrated in the DIY4U digital platform allows for easy transfer of the platform to other European industrial sectors; this will only require a replacement of product/process information and relevant models. The fablab generic approach (highlighted in section 1.3.1) will allow it to be applied to a wide range FMCG as well as soft matter products in other industrial sectors.

2.1.2.1 Creating new market opportunities for European equipment manufacturers and DIY4U users: The industrial adoption of fablabs in the FMCG sector will create a major new business opportunity for European process equipment manufacturers such as consortium partner, CDY – a robotic and automation solution provider. Access to DIY4U digital platform and fablabs will unroll a wide range of entirely new business opportunities for European particulate/liquid FMCG stakeholders (i.e. manufacturers, ingredient suppliers, innovation SMEs and entrepreneurs) to the benefit of consumers. DIY4U will facilitate the integration of new functionalities into their existing products as well as the creation of entirely new products driven by real-time analysis of customer needs/demand. Beyond the project, the DLT-enabled DIY4U platform will also allow the signing of digitised legal contracts between stakeholders with automated execution of terms; this will facilitate collaboration and hence drive innovation.

2.1.2.2 Strengthening economic growth and competitiveness of European industry: Although Europe is the world's largest economic bloc, with a GDP >€14 trillion in 2014, its growth has been progressively declining.[10] This is particularly true for the European manufacturing sector (responsible for over 80% EU exports and 80% of private research and innovation) which reach a low of 15.3% in 2014[11]. The European Commission estimates[11] that the digital revolution could generate an "additional 2.1% of GDP growth over the baseline. Digitising the European FMCG sector via adoption of DIY4U digital platform and fablabs will help in reviving the FMCG manufacturing sector, creating new jobs and safeguarding existing jobs. We estimate (section 2.2-a.1) that DIY4U will stimulate the creation of at least 36 direct jobs and over 200 indirect jobs in the EU FMCG sector within 7 years after the completion of the project. For every new manufacturing job created, 1.5-2 jobs will be created in other sectors.[7]

2.1.3 Environmental, social and other impacts of DIY4U

Reduced waste, improved resource efficiency and enhanced customer satisfaction: On-demand production of customer-centric products will reduce waste, improve resource efficiency and enhance customer satisfaction as consumers would be more likely to maximise the use of products tailored to their individual needs. For example, detergent formulations can be tailored to the specific season of the year – in the summer months having a formulation better able to tackle 'outdoor' stains such as grass may be attractive to a family with young children etc. Likewise, the formulation could be tweaked between winter and summer to reflect that people are wearing warmer clothes and hence different types of fabrics. The 'just in time' product manufacturing will relax the requirements for product stability and

[10] https://www.weforum.org/agenda/2015/03/the-digital-transformation-of-europe/ [11] R. Davies, Industry 4.0 Digitalisation for productivity and growth, European Parliamentary Research Services, 2015.

[11] Dimitri Lorenzani, Janos Varga, European Commission, The Economic Impact of Digital Structural Reforms, European Economy. Economic Papers **529**. September 2014. doi: 10.2765/70944

stabilisers which are traditionally built into products such that they can reasonably be expected to sit in a warehouse or distribution centre for many months at elevated temperature and humidity, etc. For powders, this may result in reducing the addition of anti-agents etc. whilst for liquids, it may lead to the easier addition of environmentally benign components with stability issues such as enzymes. Better performing detergents developed via the DIY4U platform and fablab can lead to a reduced wash time and hence less fabric going down the drain. ~700,000 fibres currently end up in water per domestic wash[12]. Better performing detergents will also reduce both wash temperature and water consumption.

Fablabs will allow consumers packaging choice to reduce plastics waste, enabling consumers to make ecologically driven choices at a local level. The DIY4U approach will lead to more agile response to changes in demand, further benefiting consumers and the society.

Small-scale production industries lend themselves to fewer employees per company – making management easy and economical. Closer relationships between employers and workers in small-scale industries will cause employers to look after the well-being of their employees whilst employees, on the other hand, would consider their work as theirs, leading to dispute-free and seamless operations. New DIY4U industries will also benefit from

the existing external economies (e.g. transport, electricity and communication facilities, banking and insurance services. etc) of large-scale production companies, making their operations cost-competitive.

Create employment: DIY4U will deliver digital design, innovation and manufacturing infrastructure which are currently not readily available for most entrepreneurs, start-ups and SMEs. The DIY4U approach will allow smaller companies with niche or innovative products to obtain a foothold in their respective markets in the high volume/low margin sectors where success is traditionally dependent on obtaining 'shelf space' for your products in large retailers/supermarkets. Small-scale production could be started by anyone with a small capital. This will boost the ability of new actors to become involved in the creation, making and commercialization of innovative products, creating employment. Many unemployed EU citizens would become employed in the newly developed small-scale industries.

Political and Digital Single Market impact: A key priority of EU governments is to promote the development and adoption of manufacturing and value chain innovations that enhance the competitiveness and productivity of European industries, increasing exports, reducing imports and assuring national and regional supplies. DIY4U fits this priority and will help EU governments to realise the digital transformation vision. "The industrial revolution of our time is digital. … As companies aim to scale up across the Single Market, public e-services should also meet today's needs: be digital, open and cross-border by design. The EU is the right scale for the digital times
— Andrus Ansip, Vice-President for the Digital Single Market" The digital transformation of EU business and society offers a massive growth potential for Europe. European industry can build on its strengths in advanced digital technologies and its strong presence in traditional sectors such as FMCG to capitalise on the broad range of opportunities that technologies such as advanced manufacturing, robotics, IOT, big data, artificial intelligence and DLT/blockchain offer. However, EU businesses are not yet taking full advantage of these advanced technologies or innovative business models offered by the collaborative economy.[13] DIY4U will strengthen the digital capacity of EU FMCG sector, enabling EU stakeholders to capture a considerable share in the emerging markets for the products and services of the future.

[12] https://www.bbc.co.uk/news/uk-45770358

[13] https://ec.europa.eu/growth/industry/policy/digital-transformation_en

Legal impact: The use of blockchain/DLT-based messaging system and process automation with 'smart contracts' will facilitate compliance with contractual agreements. To enable audit and reporting, DIY4U will include a reporting module that will extract messaging data from the ledger in a format that can be analyzed from a legal perspective (e.g. time sequenced communications between the designer, pulling in ingredient IP from a third party, testing this virtually using machine intelligence, and passing the final mixture to the fablab for preparation). Blockchain/DLT will also ensure consistency of transactions between parties collaborating on the creation and making of new products, tracking liability and helping to avoid IP conflicts. Moreover, using a GDPR-compliant architecture for digital security, data protection and privacy by design will contribute to better protection of consumers' privacy and personal data.

2.1.4 Barriers to achieving the expected impacts and consortium strategy:

We expect DIY4U outputs (digital platform and fablab MDFs) to be ready for market deployment shortly after the end of this project, hence minimal post-project financing will be required prior to market launch. Follow-up steps (operation, maintenance and continuous improvements of both the platform and fablab) will be funded from early revenue generated by the platform. We expect the eventual volume manufacture and deployment of fablab machines in supermarkets, homes or dedicated small-scale factories to be funded by a joint venture from relevant DIY4U partners and private investment from industry. This project will effectively de-risk such private investment by demonstrating both the demand for and practicality of the proposed DIY4U approach. The main foreseeable barriers, obstacles or framework conditions that may determine whether and to what extent the expected impacts will be achieved are summarised in table 2.1.

Table 2.1 Main Barriers to achieving DIY4U expected impacts

Barrier, obstacle or framework condition	DIY4U consortium strategy
Fears around breach of confidentiality and IP rights: Industry stakeholders might be opposed to using the open innovation platform due to fear of breach of confidentiality or disclosure of trade secrets. Barriers may therefore result from the legal framework such as the General Data Protection Regulation 2016/679/EU (GDPR) and the European Charter of Fundamental Rights.	GDPR compliance measures integrated in the design and development of the platform will ensure the security of users' data and intellectual property (IP). These will be addressed by analysing the legal obligations and monitoring practices in regulation (European Data Protection Board, former ARTICLE 29 Working Party) and standardisation (ISO, IEC, CEN, CENELEC, etc.) in support of those obligations, in the project's data protection and privacy requirements definition and compliance monitoring. Use of Blockchain/DLT will ensure data security and transparency, allowing data and information to be shared without being copied. A blockchain cyberattack requires simultaneously attack of all copies of the same shared database. We will proactively disseminate the integrated data security and IP protection measures of the platform.

Industry's reluctance to adopt disruptive business models: Large FMCG manufacturers and their suppliers might be reluctant to adopt business models that pose a major threat to their current business	DIY4U is bringing the relevant value chain together to develop innovative business models that will represent a win for all stakeholders in the innovation and manufacturing ecosystem. Increasing demand for customization/personalisation is pushing the large companies to embrace DIY4U-type approaches as differentiators to deliver greater customer satisfaction and enhance their competitiveness. Our consortium benefits from the support of a top 5 global FMCG company (P&G).
Lack of customers involvement in the codesign and manufacturing process	The use cases definition process will benchmark customers role in 3D-printed products design and manufacturing
Regulatory and standard requirements: The lack of standards for the reformulated products might present a major product safety challenge	The consortium will engage regulatory authorities (e.g. AISE) and use input from FMCG ingredient suppliers/manufacturers (e.g. P&G) to ensure its developments comply with relevant standards and regulations (e.g. Regulation (EC) No 648/2004). Only ingredients that have previously been approved to meet industry and regulatory standards will be used. The standard and regulatory information for ingredients and processing options will be displayed on the highly visible and transparent platform. The digital design and performance simulation tools will ensure that only intrinsically safe formulations can be designed and tested prior to manufacture.

2.2 Measures to maximise impact

Draft 'plan for the dissemination and exploitation of the project's results'

Dissemination, exploitation and communication of DIY4U approach and project outputs are critical for achieving the expected impacts highlighted in section 2.1, hence 30 person months in project implementation has been dedicated to dissemination, exploitation and communication activities.

2.2a Dissemination and exploitation of results

This section gives an overview of the key exploitable results (KER) and presents a draft plan for exploitation and dissemination of results which will be complemented by communication activities (section 2.2b) to promote the project, its achievements and commercialisation of its outputs. An initial plan for joint exploitation is provided, covering the business model for DIY4U and the underlying financial model.

Key exploitable results (KERs) of DIY4U: DIY4U's main KER is the establishment of the DIY4U ecosystem, which will be a self-governed and self-sustained organisation with representatives from the consortium. Three types of individual KERs are identified and shown in table 2.2a.1.

Table 2.2a.1 Key exploitable results in DIY4U

No.	KER	Partners	WP
KER1	DIY4U Digital Platform for creating new customised or personalised FMCG	All DIY4U partners	WPs 2-5, 8 &9

KER2	DIY4U Fablab MDFs for small-scale manufacture of making customised or personalised FMCG	STEF, CPI, CTM, CDY, P&G, PICOM, VTT, and value chain end-users signed up via the DIY4U platform	WPs 6-10
KER3	New customised or personalised detergent and soaps	FMCG value chain stakeholders (including entrepreneurs, start-ups, SMEs and consumers) signed up via the DIY4U platform and supported by DIY4U partners	WPs 8 & 9

Joint exploitation strategy: During the project (2019 to 2022), the DIY4U digital platform and fablab technology components will reach TRL6. In addition to each partner's individual exploitation strategy (section 2.2a.2), the consortium will develop and implement a joint strategy for transitioning the platform and fablab to marketreadiness (TRL9) and for commercialising the DIY4U platform and fablab to meet the expected impact and ensure the long-term sustainability of DIY4U. The business plan presented in section 2.2a.1 demonstrates the selfsustainability of the DIY4U ecosystem. We expect the joint exploitation of the platform to commence within a year from the completion of the project – the DIY4U platform will offer DIY digital design and performance simulation services to any non-expert wishing to create a new formulated detergent and soap product. STEF and CPI individual exploitation strategy for the MDFs will complement the joint exploitation of the DIY4U platform by providing open access DIY manufacturing facilities for platform users to manufacture the products designed on the platform. The joint exploitation strategy for the fablab will focus on transitioning it from TRL6 to TRL9 within 2 years of completing this project. We expect to deploy DIY4U fablabs across the EU market by the end of 2024, working in partnership with supermarkets and/or innovative SMEs wishing to offer DIY4U small-scale manufacturing service.

The project's exploitation will be governed by DESCA 2020 Model Consortium Agreement detailing background knowledge/IP, foreground IP, rights, obligations and relationships between them.

2.2-a.1 Business plan

This subsection analyses the relevant market segments, provides a preliminary business model canvas, performs SWOT and competition analysis and provides a revenue forecast for the DIY4U ecosystem.

DIY4U target markets: Although DIY4U is potentially targeted at the whole soft matter FMCG market (including detergent and soap, cosmetics, food, gym supplements, agrochemicals, polymer gels, pastes, foams, coatings, adhesives, etc), the immediate target market is the detergents and soaps market. The global detergents and soaps market was worth €86.21 billion in 2016[14] and expected to reach €180.55 billion by 2025,[15] driven by increasing health awareness, rising disposable income and growing sales of washing machines in both developing and developed regions[16]. Product innovation and new product launch (a key priority of DIY4U) is also a major factor expected to drive the market growth. Vendors are expanding their businesses by building manufacturing facilities in developing economies such as China and India. Some of the key players catering the market growth are Ecolab Inc.; Unilever; Church & Dwight Co.; **Procter & Gamble (consortium partner)**; Henkel AG & Co. KGaA;
Colgate-Palmolive Company; Lion Corp.; Reckitt Benckiser Group PLC; The Clorox Company; and Amway. The

[14] Grand View Research, Global Soap and Detergent Market Size, Industry Report, 2018-2025, January 2018, Report ID: GVR-2-68038-299-0

[15] https://www.giiresearch.com/report/grvi615535-soap-detergent-market-size-report-by-product.html

[16] Grand View Research, Global Soap and Detergent Market Size, Industry Report, 2018-2025, January 2018, Report ID: GVR-2-68038-299-0

market is witnessing globalization, with global vendors branching out to increase market share. North America dominates the market with the highest revenue share whilst Asia Pacific is expected to witness the highest growth to 2025. Although the European market is projected to grow at a significant rate to 2025, the European sector needs to adopt disruptive approaches (like DIY4U) to outperform North American and Asian competitors and to reshore manufacturing in the EU. The competitive advantages and the collaboration/business generation capability provided by DIY4U to EU FMGC stakeholders will enable them to capture additional 5-10% share of the soft matter global FMCG market within 10 years (by 2035) from DIY4U launch, generating tens of billions€ in annual revenues.

Initial business model canvas for DIY4U: The economic sustainability of DIY4U is established by the Business Model Canvas analysis presented in the table below. It analyses the main parameters affecting DIY4U's financial performance including the key partners, activities, and resources, the value proposition to its customers, customer relationship and distribution channels, customer segments, the cost structure, and the revenue streams. The Business Model Canvas can be applied to DIY4U as a whole, or to individual services within DIY4U (i.e. digital platform or fablab MDFs). Table 2.2a.2 shows our preliminary Business Model Canvas for the DIY4U ecosystem; further analysis of individual services will be developed during the project. *Table 2.2a.2 DIY4U Preliminary business model canvas*

The Business Model Canvas designed for DIY4U v.1.0				
KEY PARTNERS - Partner R&D organisations - SME technology developers - Particulate and liquid FMCG value chain including ingredient suppliers, manufacturers, distributors/retailers and consumers - Governmental agencies - Networking partners - Investors and seed funds - European Commission	KEY ACTIVITIES - Research and development - Testing and validation - Customer tailored workflow development - Technology and market analysis - Access to finance - Business development - Technical support on DIY design and smallscale manufacturing - Community building - Learning and sharing best practices - Training - Workshops and seminars	VALUE PROPOSITION - Mitigation of R&D costs and risks for-personalised FMCG product development and small-scale manufacturing of customised products - Access to state-of-the-art digital infrastructure and smallscale manufacturing equipment - Access to open source product data exchange and standard representations of products and processes that ensure the compatibility of modelling and simulation with different process information systems - Enabling sharing and analysis of relevant data and demands from users - Enabling collaborative engineering in	CUSTOMER RELATIONSHIP -Web-based DIY4U platform National/regional DIY4U contact points - Personal relationships - Newsletters - Project follow-up - Workshops - Dissemination material	CUSTOMER SEGMENTS - Innovative SMEs and large firms in the particulate and liquid FMCG sectors. - Entrepreneurs and start-ups with no technical know-how - FMCG retailers - FMCG Consumers - Knowledge transfer and business consulting companies - Industries within material science

	KEY RESOURCES - Expert personnel - Infrastructures (fablab MDFs) - Knowledge and IP - The DIY4U platform - Network of experts - Professional/scientific societies - Outreach activities	the production network, allowing all actors to propose innovations - Enabling 'plug and produce' of personalised/ customized FMCG - Access to knowhow and highly skilled personnel - Improving efficiency and effectiveness of product development, customisation & small-scale manufacturing -Increasing product variety and personalisation for higher customer satisfaction and loyalty - Early access to R&D activities and trends - Identifying technological and market gaps - Vision creation - Professional networking - Creating pan-EU cooperation	CHANNELS - The DIY4U web-based platform - Cascade network - Partner's networks -Affiliated organisations	

COST STRUCTURE - Infrastructure (purchase, maintenance, depreciation) - Fixed costs (rent, utilities, etc.) - Staff (technical, administrative, business development) - Events organisation - Networking	REVENUE STREAMS - DIY digital design services - MDF services - Fablab replication (DIY small-scale manufacturing services) - Private and public funding for future R&D projects at MDFs

DIY4U SWOT Analysis: DIY4U aims to penetrate the target markets highlighted earlier by promoting the value proposition (table 2.2a.2) of the cost-effective and readily available state-of-the-art DIY4U platform and fablab MDFs. DIY4U will benefit European industry, particularly SMEs lacking adequate knowhow and design/manufacturing infrastructure. EU SMEs and large companies can take advantage of the technical/business insights and capabilities provided by DIY4U to innovate and enhance their competitiveness in the target global FMCG market segments.

The analysis of the overall strategic position of the DIY4U ecosystem/business and its environment is provided in a SWOT analysis (table 2.2a.3). The SWOT analysis aims to identify the strategies that will create a DIY4U specific business model that will best align its resources and capabilities to the requirements of the environment in which the DIY4U ecosystem/joint venture company will operate.

Table 2.2a.3 DIY4U SWOT analysis

Strengths
Strong consortium with an existing commercial presence in the FMCG target markets – P&G is amongst the top 5 global FMCG companies with a strong network of retailers and customer base that will provide the initia customer base for DIY4U; leading French retailers accessed via PICOM's retail innovation hub; strong commercialisation and go-to-market capabilities; and unique digital platform and fablab with strong user value propositions (table 2.2-a.2).
Weaknesses
DIY4U platform and fablab services are not known in the market yet. This will be addressed by development and promotion of DIY4U brand via digital marketing and other channels (section 2.2a.3); and by leveraging the strong network of industrial consortium partners to promote DIY4U visibility in the target market.
Opportunities
Competing regions could develop and bring similar platforms and fablabs to market over time. DIY4U will benefi from early-entrant advantage, capturing and consolidating a considerable customer base ahead of competitors, and innovative DLT-based community governance. Continuous improvement of the platform's capabilities, products and services based on real-time user feedback and integration of new innovations developed by EU SMEs, RTOs and universities will allow DIY4U to remain competitive. Lack of standards for digital DIY formulation design and DIY small-scale formulation manufacturing standards could represent a major barrier. This will be addressed by engaging regulatory bodies and using their input together with input from relevant industry stakeholders to develop software mechanism for ensuring designed and manufactured products are compliant with relevant safety and regulatory standards.
Threats
The main threats are the resistance to change and the fear of unknown DIY products. Also, the difficulty to certificate customised/personalised products (e.g. CE labelling) and the lack of rules and regulation that manage and supervise to the DIY sectors are important challenges for this project. The analysis of data in a wrong way can damage the reputation of fablabs. There is a distinct possibility that blockchain-based value transfer will be outlawed or heavily taxed by G20 nations to protect the banking sector, thus abolishing the opportunity of using this technology for automating international exchange of goods-for-payments. Finally, cyberattacks agains participants and the platform still remain a serious risk for the stored private data.

Competitor Analysis: There is currently no open innovation digital platform and fablab offering the proposed DIY4U services and value proposition. The nearest competing offering (detailed in section 1.4) is only applicable to paints rather than to all soft matter FMCG as for DIY4U. Paint customization services fall short of the capabilities (e.g. DIY formulation design, DIY automated small-scale manufacturing, open innovation, training, real-time demand tracking and respond, etc) afforded by the DIY4U holistic approach.

DIY4U turnover capacity: To ensure the continuity and the autonomy of DIY4U post-project, the DIY4U platform will generate revenues based on the following four main profitable activities:

Fablab implementation: DIY4U joint venture will found its own Fablabs in order to manufacture and sell products without intermediate to the finale costumer (B2B solution).

The Fablab replication and the Platform visibility will help to increase gradually the Fablabs customers. The consortium used the following assumptions to evaluate the evolution of the profits per year. The initial number of DIY4U consortium Fablabs are 10 dispatched in whole EU. The Fablabs customers number is percentage of EU households which is considered 128 million with a total consumption of 8€/month for shampoo, detergent and soap products. This percentage will start from 0.1% the first year to 1% the fifth year. The Figure 2.2a.1. Shows the Fablabs incomes against theirs fees. The Fablabs number will increase according to the Fablabs incomes and customers number.

Pay Per Click (PPC)-Advertising: model of web marketing in which advertisers (e.g. Google Adsense) pay a fee each time one of their ads is clicked. Essentially, it's a way of buying visits to the DIY4U platform.

DIY4U cloud-based services: the DIY4U platform will propose simulation tools to design and test formulations. These web services will be based on upselling method and recommend three offers (basic, silver and gold categories) for validated customers.
Extra commissions: For each order placed and delivered via the DIY4U platform, a conservative 5% commission will be added to the total amount of the order.
Fablab replication: DIY4U joint venture will manufacture and sell or lease equipment to supermarkets and SMEs in return for a percentage of sales of products manufactured using Fablabs.

The turnover estimation strategy is based essentially on the number of visitors. The consortium used the following assumptions to evaluate the evolution of the profits per quarter:

- The initial number of visitors per day of the DIY4U platform is 150
- For each relevant 1000 visitors, it will be assumed to provide 20 significant clicks generating approximately 2€. The growth average of the platform views per year is 50%
- It is assumed that 0,1% of the visitors will validate their subscription of formulation services which the average costs is 20€ per month.
- The consortium supposes that 20% of formulation simulation will be concluded by new orders and the average amount of the commissions is about 100€.

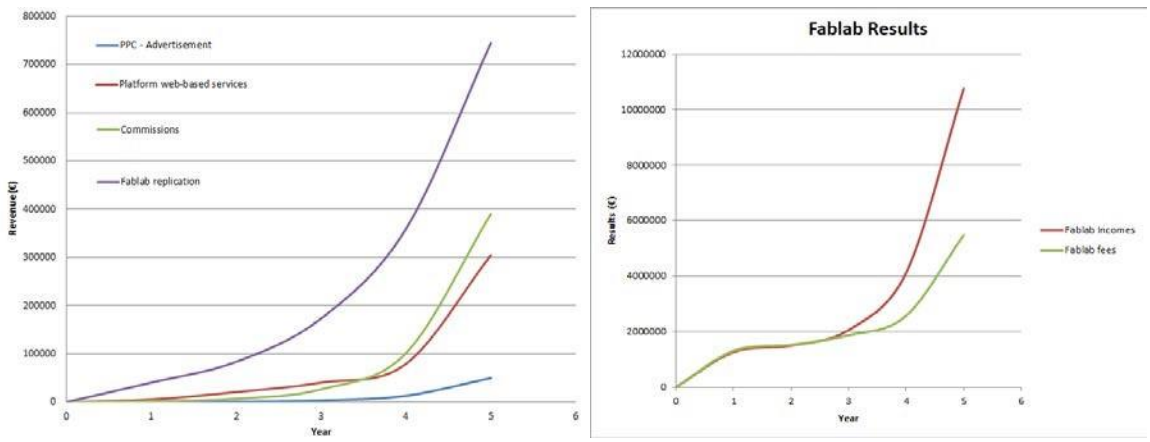


Figure 2.2a.1: Estimation of revenues during 5 years

Figure 2.2a.1 illustrates that the fablab replication services of formulation represent the most important source of revenue. Therefore, this project will accord particular importance to develop these services. Table 2.2a.4 shows the estimated turnover for 5 years post-launch. In year 5, the DIY4U platform will generate ~7.27M€, creating 36 direct durable jobs (1 job = 200K€ revenue) and training students in the IT and digital manufacturing field.

Table 2.2a.4 the turnover estimation in 5 years

Turnover estimation	*Year 1*	*Year 2*	*Year 3*	*Year 4*	*Year 5*
Benefits	*1,313,313.95 €*	*1,611,007.56 €*	*2,340,406.45 €*	*4,894,390.28 €*	*1,3105,774.36 €*
Costs	*1,555,067.14 €*	*1,771,432.08 €*	*2,114,950.47 €*	*2,914,089.19 €*	*5,835,371.44 €*
Results	*-241,753.2 €*	*-160,424.52 €*	*225,455.97 €*	*1,980,301.09 €*	*7,270,402.92 €*

Cumulative profit	*-241,753.2 €*	*-402, 177.72 €*	*-176,721.75 €*	*1,803,579.35 €*	*9,073,982.26 €*

Figure 2.2a.2 shows the evolution of the benefits, costs and the economic results of the DIY4U platform during 5 years. The figure shows that DIY4U will be viable, with profit of 225K€ in year 3 and the break-even point after 4 years (cumulative gross profit of 1,8M€ by the end of year 4).

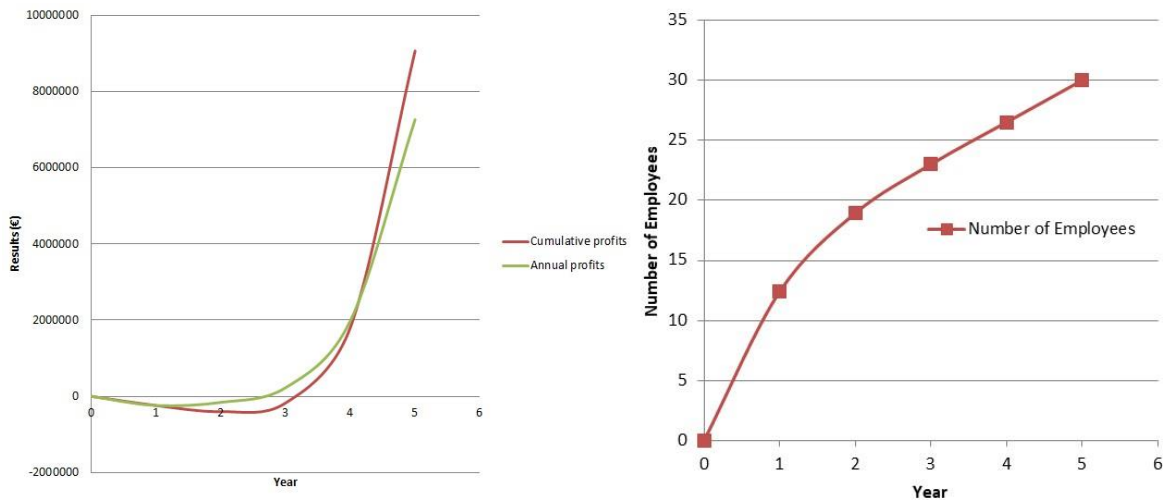


Figure 2.2a.2 turnover and direct job creation estimation

In addition to these monetised results, the DIY4U will create non-monetary benefits and indirect outcomes such as skilled training, quality assurance, protection of environment and health safety.

2.2-a.2 Individual exploitation strategy

Partner exploitation plans are aligned with the principles of the DIY4U business plan presented in the previous section. All individual exploitation plans will be continually monitored by the Exploitation Leader. Table 2.2a.4 summarises the individual exploitation plans of the DIY4U consortium partners

Table 2.2a.4 Exploitation plan for DIY4U consortium partners

Partner	Exploitation plan
SME Partner	SME partners will enhance their competitiveness and growth by developing and exploiting innovative components of DIY4U as detailed below: • IRIS will gain knowhow relating to the decentralized architecture of the platform and implementation of hybrid blockchain technologies that foster private and public collaboration. In addition IRIS will diversify its portfolio of digital services to the industry by becoming a key player FMCG collaborative production industries of the future, fostering the company's growth. • EFF will gain knowhow relating to the decentralized architecture of the platform and implementation of hybrid blockchain technologies that foster private and public collaboration. This will enable EFF to enhance its competitiveness as well as become a key player in the FMCG collaborative production industries of the future, fostering the company's growth. • RDI will acquire new skills in data analytics related to FMCG sector and improve its know-how in knowledge base elaboration, enabling it to become a key player in the future collaborative
	production industries. RDI will enhance its capacities to develop industrial open technologies for the FMCG sector and other sectors. •

	complete p on the detergent/surfactant and FMCG industries to provide a e areas of process engineering. DSC will gain additional future growth. rvices to the prior industries which would foster the company's ● CDY will be at the centre of fablab become a key stakeholder in the FM e joint exploitation strategy earlier business opportunity for CDY, allowing it to grow its business
FMCG Industry partners	Industrial partner (P&G) will increase its market competitiveness in the FMGC sector by ensuring defined activities as follows: ● Supplying base materials for DIY4U manufacture worldwide ● Applying learnings (technological and business model) from the projects to other innovation programmes within the company ● Upskilling staff on small scale digital manufacture
RTOs	RTOs expect to contribute in the development and manufacture of a variety of new innovative ● STEF will apply the knowledge developed through the project in other business sectors; Feed pellets production, fertiliser formulations, etc. STEF will use Fablab facilities to create new industrial and public funded projects. The fablab facilities will be useful in STEF's activities within digitalisation of process industries. ● CPI will develop digital formulation design know-how and small scale formulation manufacturing capability, enabling it to expand its service offerings to include small-scale automated capabilities for the wider materials formulation and manufacturing community. ● Digital Catapult (DCC) will apply the resulting DLT technologies in other R&D manufacturing programmes, specifically looking at connected supply chains. Digital Catapult will be able to capture real-world data from DIY4U and showcase to traditional industry to prove the value proposition of this technology and this will enable the growth of DLT Field Labs programme. The experience gained in this project will also contribute to further develop consulting services to educate start-ups and scale ups understand business models and user experience with distributed systems. ● PICOM will gain better knowledge of customisation business models and will be able to transfer this knowledge to other types of goods; helping retailers and brands understand the opportunity and feasibility of such new business. ● CTM will improve their knowledge in the development of fablabs and open innovation technologies that will be transferred to local stakeholders of the FMCG sector through the dissemination program, which will contribute to improve the innovative capacity and competitiveness of the region. Furthermore, this project will open the possibility to make new contacts and develop a networking with other European partners around the results of the project and for taking advantage of them for other synergistic projects. ● VTT will apply the PAT know-how created in DIY4U to different application areas, such as pharmaceutical industry and food industry but also to develop concepts for consumer-based chemical analysis. VTT will use the results to prepare new industrial and public-funded projects. ● STL will further build on its knowhow in legal data privacy requirements analysis and GDPR compliance for such collaborative design and production, enhancing its service offering to a wider industry base.

2.2-a.3 Dissemination and exploitation of results

DIY4U initial plan for exploitation and dissemination of results (PEDR) has been developed and will be regularly updated during the project. The updated PEDR will be included in the project's semi-annual periodic progress reports. The plan sets out details of activities to be undertaken, focusing on the following target stakeholder groups:

Group 1: *Detergent and soap end-users:* e.g. consumers, cleaning service providers and cleaning solution developers (e.g. Deb Ltd, Kiehl Group, Laboratoires Ceetal, Mareva)
Group 2: *EU Innovators (SMEs and Entrepreneurs) of detergent and soaps and other soft matter FMCG products:* e.g. Reckitt Benckiser, People Against Dirty, AGMA, Wolf Formulations,
Group 3: EU *Manufacturers of detergents and soaps and other soft matter FMCG:* e.g. Unilever, ReAgent, Amega Sciences, Autosmart, Christeyns, Ecochem Srl, Jabones Pardo S.A.,
Group 4: *EU detergent and soap Ingredient Suppliers:* Grupo Industrial Crimidesa. BASF, AlindaVelco S.A., CP Kelco, Solvay, EVONIK, PQ Corporation, Kmira, INEOS, Dow Chemicals, Novozymes, GENeco, SMEs developing novel ingredients (e.g. Microcapture, Filda London)
Group 5*: Distributors and Retailers of FMCG:* e.g. Lidl, Asda, Tesco, Aldi, Sainsbury, Auchan, Carrefour, Netto,
Hyper U, SPAR, Coop Mega, Meny, Kesko
Group 6: *EU ICT solution developers & other fablab developers:* e.g. Fablab Valencia, FabLab Pavia
Group 7: *Primary influential bodies/industry associations:* e.g. FoodDrinkEurope, Independent Retail Europe, European Commission, European Environment Agency, United Nations Environment Programme (UNEP, The Intergovernmental Panel on Climate Change (IPCC) and policy makers etc.
Group 8: *Regulatory/standardisation/legal counsel:* e.g. AISE, SGS, IP law firms, digital transformation and AI advisors, cybersecurity auditors, consultants for privacy, identity management and blockchain, etc.
Group 9: *Investors:* Banks, venture capitalist, public and regional fund providers
Group 10: *Other:* Academia and public

Dissemination strategy and measures: Our dissemination strategy will focus on raising awareness of DIY4U and its user benefits. This will be achieved by the development and publishing of case studies; newsletters; magazine articles and video/animation material. Our approach for disseminating the results of DIY4U will cover: (1) Publishing scientific and technical results from WPs in journals and at conferences. This activity will allow the scientific community to benefit from outputs that have potential to inspire and lead to future scientific research. (2) Trade press and trade fair dissemination to increase awareness of DIY4U amongst our target user base. This will include press releases to European environment agencies and relevant regulatory bodies.

We have planned to publish at least 5 papers within the project lifespan. These papers will focus on topics including: cost, resource and energy efficient potential of the DIY4U approach. Transferrable outputs relating to formulation modelling and simulation, and AI/big data.

We aim to achieve 100,000 viewers of the project website by using digital marketing strategies such as search engine optimisation (SEO) and quarterly reviewing/fine-tuning our strategies.

The European Commission will also be a key dissemination route for DIY4U. The promotion of further collaborative Research and Innovation Action (in addition to the technology developed) is crucial in gaining the optimum value from pan-EU collaborative projects such as DIY4U. We will provide the EC with progress updates suitable for use in promoting Research and Innovation Action to European

companies. We will also share experience with other EU programme members and potential members whenever possible (e.g. at EFFRA events). A list all dissemination measures to be completed within the project to reach all DIY4U 'Target Audiences' is provided in Table 2.2a.5.

Table 2.2a.5 Dissemination objectives and roles of partners

Target	Dissemination Activities	Partner Responsible	Dissemination objectives
All target Stakeholders **Group 1** to **Group 10**	Project Website	STEF	Ready in Month 1, updated bi-monthly, 100,000 visitors during the project.
	Project videos	CTM	One presentation and one video to disseminate the platform and fablab MDFs to end-users.
	Brochures, posters	PIC/CPI/ STEF/ VTT	Communication templates; brochure & poster
	Facebook, Twitter, LinkedIn Groups	ALL	Over 500 Twitter followers, over 500 likes on Facebook; over 250 LinkedIn members.
	DIY4U newsletters	PIC/CPI/ STEF/IRIS/DCC	One newsletter every 6 months
	DIY4U community	P&G/PIC/DCC	At least 50 external organisations engaged by project end.
	Press releases	ALL	1 press release per Milestone to be distributed by all partners
	Media appearances	ALL	1 newspaper article or media appearance on national channels per partner
	Scientific Conferences	ALL	1 conference per technical partner to present results; Active distribution of at least 250 leaflets
	Scientific Journals	STEF/CPI/CTM/VTT/DCC	1 Article and publication per technical partner.
Users of DIY4U platform and fablab MDFs **Group 1** to **Group 6**	Dissemination through Events, Workshops, etc		
	Open Innovation Competition	DCC/PIC	Identification and engagement with EU ecosystem of SMEs that want to collaborate and underpin business opportunities
	EU/National events in Manufacturing Domain	PIC/CPI/VTT/ STEF	Attend 10 EU/National events to promote the DIY4U Platform and fablab MDFs.
	Events	CPI/STEF	Promote platform and MDFs at these events. Each technical partner will have a target to present DIY4U in at least one presentation slot at one event.
	Workshops, Events set by consortium	ALL	
	Collaboration with national or European Networks or Initiatives		
	EFFRA	ALL	At least one talk at EFFRA.
	euRobotics	CTM/CDY	At least one presentation at euRobotics
	EU Platforms like EuMAT, ManuFUTURE,	IRIS/DCC/EFF/ RDI'UP/DSC	Disseminate projects results in those Platforms
	EU Initiatives, e.g., TagItSmart,AceForm 4.0	PIC/CPI/ALL	Disseminate projects results within these Initiatives

Exploitation plan, strategy and measures: The consortium will seek to protect the overall DIY4U approach by patenting the DIY4U platform as a unique combination of hardware, algorithms and software and DIY4U fablab as a novel smart/digital small-scale manufacturing machine for soft matter formulated products. The consortium has followed the EC's "Horizon 2020 - A Guide to IP Management" and prepared a draft consortium agreement encompassing ownership and exploitation rights. The intellectual property rights (IPR) arising from this project will be the property of the consortium member that has created it. DIY4U's novel platform and fablab for creating and manufacturing formulated products will be protected via patents (DIY4U approach), design rights

(digital platform, fablab), trademark (digital platform and fablab), trade secrets (knowhow) and copyright (software integrated in both platform and fablab). The consortium agreement will include terms for managing ownership and access to key knowledge (IPR, research data, etc.), allowing the partners to collectively and individually pursue market opportunities arising from the project's results. The consortium will leverage STEF and CPI extensive knowhow and experience in innovation management together with STL's legal expertise to support third-party in optimally protecting and exploiting third-party IPR created during the project. **DIY4U draft exploitation plan** covers the ten years period beginning from the project implementation phase. The DIY4U exploitation strategy has been divided into two major phases:

Years 1-3 (2019-2022) DIY4U project implementation: This phase will cover the development of the DIY4U technology components to TRL6. Alongside the technology development, we will also focus on: IP-protecting the exploitable outcomes of the project; policy and strategy development and awareness build-up activities so that we can achieve the target DIY4U membership community. We anticipate that at least 1,000 potential users of DIY4U platform and fablab MDFs from stakeholder groups 1-6 will participate in the creation and making of customized detergents and soaps.

Years 1-7 (2022-2029): During this phase, we will spin out a company named 'DIY4U Ltd' to commercialise DIY4U platform and fablab. We aim to achieve a 5,000-strong DIY4U membership community and to drive 510% of global formulation and sales of customised FMCG through DIY4U fablabs and platform by 2035. To achieve this, we will continue R&D, dissemination and marketing activities. We plan to spend 10% and 20% of revenue generated from the platform on marketing and R&D activities, respectively. We will employ the following DIY4U branding strategy during marketing.

Table 2.2a.6 DIY4U Branding Strategy

DIY4U Vision
Open innovation digital platform and small-scale manufacturing machines that support customer-driven production all around Europe; enabling experimentation to respond to rapid demand changes and shorter time to market - improvement of the codesign and co-development capabilities towards a reduction of development costs of new products and services; increasing product variety and personalisation for higher customer satisfaction and loyalty; boosting European competitiveness and market share in FMCG sector.
DIY4U Purpose

Help EU FMCG sector to adopt open innovation and fablab approaches that will lead to new products with new functionalities and shorter time to market; enabling the creation of collaborative production liaising companies, including especially SMEs, that will use these new approaches to unroll a wide range of entirely new business opportunities for the benefit of consumers. DIY4U will develop and commercialise an open innovation digital platform and small-scale digitally-enabled manufacturing machines (fablabs) for the creation and making of customised or personalised FMCG (particulates and liquids).

The DIY4U digital platform brings together the EU soft matter FMCG value chain, including manufacturers, suppliers and consumers, and provides them with tools to rapidly and collaboratively design and simulate the performance of new consumer-centric products. It helps users create personalised/customised products matching their needs and selects the right processing conditions for makers, hence streamlining product development time and cost. The platform will enable:

- Capturing of knowledge and ideas of design and engineering coming from different and even new actors; allowing all actors to propose innovative solutions;
- Sharing and real-time analyses of relevant data and demands from FMCG consumers to inform product design. Customer feedback, real-time use feedback (anonymised) and associated options;
- 'Plug and produce' formulation creation facilitated by open source product data exchange and standard representations of products and processes that ensure the compatibility of modelling and simulation with different process information systems.
- Scalability with respect to technological development and soft matter manufacturing application domains;
- Checking of product compliance to industry/safety standards and regulatory compliance, as well as security requirements.

Fablabs will be used for rapidly making and characterising products designed via the digital platform. The integration of PAT and relevant automation into the fablabs will ensure 'right first time manufacturing', limiting failures/downtime, enhancing energy efficiency, reducing operation cost and increasing productivity. MDFs (hosting the developed fablabs) will provide open access trial, allowing industry to build confidence in the approach and hence de-risking the adoption and replication of DIY4U fablab.

DIY4U Values

High quality, reliability and customer satisfaction based on reliable performance

Brand implementation

✓ Build consistent brand to reinforce **DIY4U** image including logo, marketing materials and style guides

✓ Reflect **DIY4U** aims and vision in hiring and training staff

✓ Develop DIY4U platform and fablab support and customer relations to implement **DIY4U** vision

✓ Reinforce awareness of **DIY4U** brand via promotion and marketing material including targeted digital marketing campaigns

2.2-a.4 Strategy for management of research data and knowledge management

2.2-a.4.1 Project data management

A data management plan (DMP) will be developed to handle data (both exploitable and non-confidential) generated during the project. The DMP will outline the principles and processes for data collection, organisation, management, storage, security, analysis and sharing during the project. The DMP will be essential in ensuring data protection and confidentiality where relevant. A first draft will be created by M6 and updated periodically during the project. We will comply with the ISO standard ISO 15489-1:2001 Information and documentation -- Records management and EU General Data Protection Regulation (GDPR). Data created within the project will be treated as confidential. Confidential data (designs, specifications, test methods, results, regulatory/standards, product and process documentation) will be shared and used on the DIY4U platform only through authorization from the owners of the data. Suppliers of products/services on the platform will be asked to provide consent for such data to be used for the purpose for which the platform is designed during sign up and provision of the information. The use of blockchain/DLT will ensure security, attestation, and granular transparency in the base messaging/data passing layer.

DIY4U will use DLT-based records to cryptographically prove to its users that data provided has not been tampered with, intentionally or accidentally. This will be applied in DIY4U to provide trustworthy information about product and service delivery. Non-confidential information generated by the DIY4U consortium will be shared more widely via dissemination and communication activities detailed in this section. Provided/stored data will only be used for the purpose(s) for which the data was provided. DIY4U cost for curation and preservation of data generated by the partners during the project will be covered as part of the grant received, data preservation costs beyond the lifespan of the project will be covered by the consortium partners with part of the revenue generated by the platform and fablab MDFs.
Data Storage: Research Data Storage Facility (RDSF) will be used to store software and training data during the development phase of the project. We will ensure Backup procedures are robust (overnight backup, copies held remotely on server (local and cloud)) and secured access is in place. Once live, the distributed architecture of the DIY4U system provides its own guarantees for redundancy, backup and security of the software and data within the system.
Metadata: No special metadata/cataloguing difficulty is expected during the project. Simple file/folder naming will be used to locate each piece of data as it is needed. A spreadsheet will be used to log details. Extra documentation will be created for dissemination purposes. In addition to serving as a data storage facility for the duration of the project, IRIS will undertake to preserve each of the digital outputs mentioned above, for three years beyond the end of the project on the Research Data Storage Facility, undertaking refreshment of storage media as required. These materials will be available on request. In order to make the data easily discoverable, digital object identifier will be used. The data generated will be under license protection and will be made available freely for education purpose. Enough data will be provided so that peers can validate the authenticity of the data and results. The data will be made usable beyond the original purpose if deemed necessary. The data format will be such that it can be easily converted to other formats so that interoperability is ensured. Once live, the distributed architecture of DIY4U will provide its own guarantees for the security, persistence, interoperability and discoverability of metadata.

2.2-a.4.2 Knowledge management and open access strategy

To ensure that all new knowledge and intellectual property generated by the project is managed properly and adequately protected, the consortium has developed a preliminary IPR strategy. This strategy will be further developed in an intellectual property rights agreement to complement the relevant articles of the Consortium Agreement which will be used as the starting point for the establishment of the entity to operate the DIY4U platform and fablab commercialisation after the project. The partners will adopt a proactive policy pursuing all possible patents and licensing opportunities arising from this project. Licensing of the technology to third parties outside the consortium will be considered by the Exploitation & Dissemination Manager. There are no existing or future commercial agreements between any of the partners that will impose limitations on the patenting and exploitation of results. All the foregoing clauses will be stated and formalized in the Collaboration Agreement (CA). The background (prior contract) IPR will remain the property of the originating organisation but will where appropriate be made available for use by the consortium during the project to further develop DIY4U technology components. During the project implementation period, to enhance the credibility and acceptability of the project results, we will follow the relevant standards related to the development of DIY4U platform and fablab.

A draft IPR agreement that comprehensively addresses the IPR issues in the project will be produced and agreed by consortium members. The IPR framework for the operation of the DIY4U platform and fablab MDFs with external users will also be agreed by the consortium and adapted according to the needs of the users. IPR agreement will be the responsibility of the Exploitation & Dissemination Manager. The agreement will be a combined plan with a clear, reliable and robust policy regarding the

protection of IP, including the mechanisms for policing and defence of any patents and other IPR. The following tasks will be led by the appointed Exploitation & Dissemination Manager, with support from the rest of the consortium.

- A thorough worldwide patent search to identify any prior disclosures, complementary IPR or other relevant data, assessing novelty and patentability (an update of the work performed during the proposal preparation).
- Peer review of the knowledge or technology both within the consortium and (under the terms of a concluded confidentiality agreement) from key-opinion leaders in the relevant fields, assessing novel and inventiveness.
- Drafting and filing of a robust patent application to establish a priority date.
- Development of content and claims of the patent application, along with ongoing reviews of any other work in the same/similar fields, to ensure uniqueness and maximum coverage.
- Periodic review of the patent application by the consortium to ensure relevance, value and commitment to defend.
- Where appropriate, filing of a full patent application and pursuit to grant of the patent.

The overall strategy for IPR will be coordinated at a Steering Committee level by the Exploitation & Dissemination Manager who will report his or her activities on a six-monthly basis unless a matter of urgency arises. This will include specific provisions for ownership of the IPR by the consortium members and the granting of licenses.

The consortium will also set the framework for the generation, use and protection of IPR belonging to the users of the DIY platform and fablab. It is anticipated that each use case will have to be evaluated individually to find the right IPR solution, for example there may be particular expectations with regard to IPR depending on the source of finance. The basic principle will be that the results are owned by the entity generating them. Under commercial operation, alternative arrangements may be agreed. To the extent that end-users are supported by European funds then the results will be owned by the beneficiary generating them and will be exploited and disseminated for the benefit of the EU.

2.2-b Communication activities

The project's communication activities will follow a cohesive line built around an easy-identifiable digital identity, centred around the project's web site and the digital platform in connection with relevant existing initiatives and projects through social media mainly: Facebook, Twitter, LinkedIn, Newsletters and Press releases.

A Media attention will be sought at all levels and will include the printed press, radio, television, specialised press magazines and the digital press. Moreover, project brochures, flyers, webinars, virtual info-days and electronic newsletter will be deployed on a regular frequency to inform news and updates about the project and clusters activities.

Within this cohesive communication framework, three aspects will receive specific attention. First, it creates a common purpose between the project members and the target people who participate at the trainings and the workshops; thus, also contributing to the homogenization of the various activities that are carried out in WP11. Second, it is also a means to let the project be widely known and attractive.

Based on a participatory approach, the project's communication strategy is focused on optimizing the communication means between consumers, makers and other actors, to facilitate the circulation of information, discussion rooms of mutual interest topics, knowledge exchange and opportunities for innovative DIY4U products. Communication is a fundamental element in the process of understanding, knowledge sharing and innovation creation.

Finally, any communication activity related to this project (including in electronic form, via social media, website etc.) and major results funded by the grant will:

- Display clearly the European Commission (EC) and DIY4U logos,
- Include the following mention: "This project has received funding from the European Union's Horizon

2020 under grant agreement No 870148"

Table 2.2b List of communication strategy and KPIs for each

Communication strategy	Target group	Means of communication	Communication level	When	Responsible	Performance indicators
Objectives and basic information on project, public deliverables	FMCG Industry, Policy Makers, Consumers and stakeholders	DIY4U website	International	M1-M24	SINTEF	5000 visits and 100 downloads per public deliverable one year after the project's end
Project's activities	DIY4U Users	Newsletter, social media (Facebook page, linked-in groups, twitter hashtag)	International	M1-M30	All partners	300 registered mails, 500 likes on Facebook page, 12 linked-in posts and 24 twitter tweets per year
Achievements	General Public, Investors	Two YouTube promotional videos, leaflets	Europe	M3, M30	CTM	250 views per video in 12 months from release
Lessons learnt	Advisors	Best practices handbook	Europe	M30	P&G, CDY & PIC	50after the project's end downloads in one year
Publications	Researchers, Research Centres	High impact journals (at least one publication from each WPs; 3, 4, 5, 6 and 8)	International	M10, M20, M24 and M34	All partners	5 publications and 5 citations in three years

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

The proposed work-plan consists of eleven work-packages (WPs) as illustrated in figure 3.1a. The work-plan confirms the sustainable strategy adopted by DIY4U consortium. These 11 WPs are based on the continuous improvement approach and show the transfer between the open digital platform and the Fablabs to manufacture customized products while respecting rules and standards and ensuring security. WP1 will ensure the successful management and implementation of the project whilst WP11 will disseminate the outcomes and maximise the exploitation of the expected results.

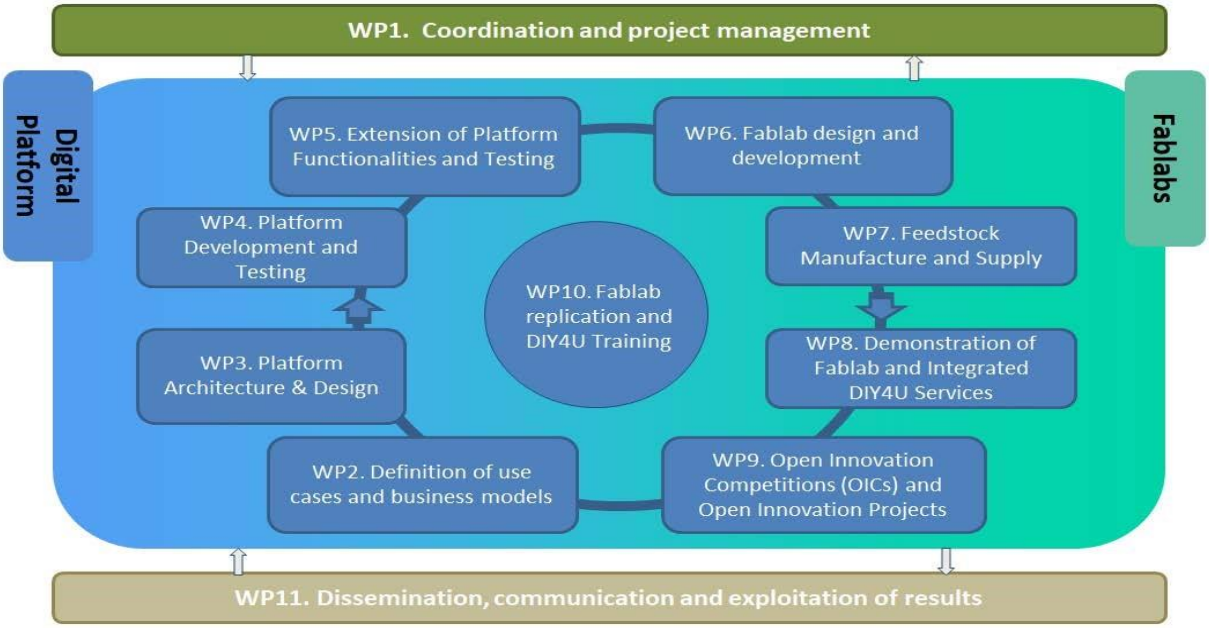
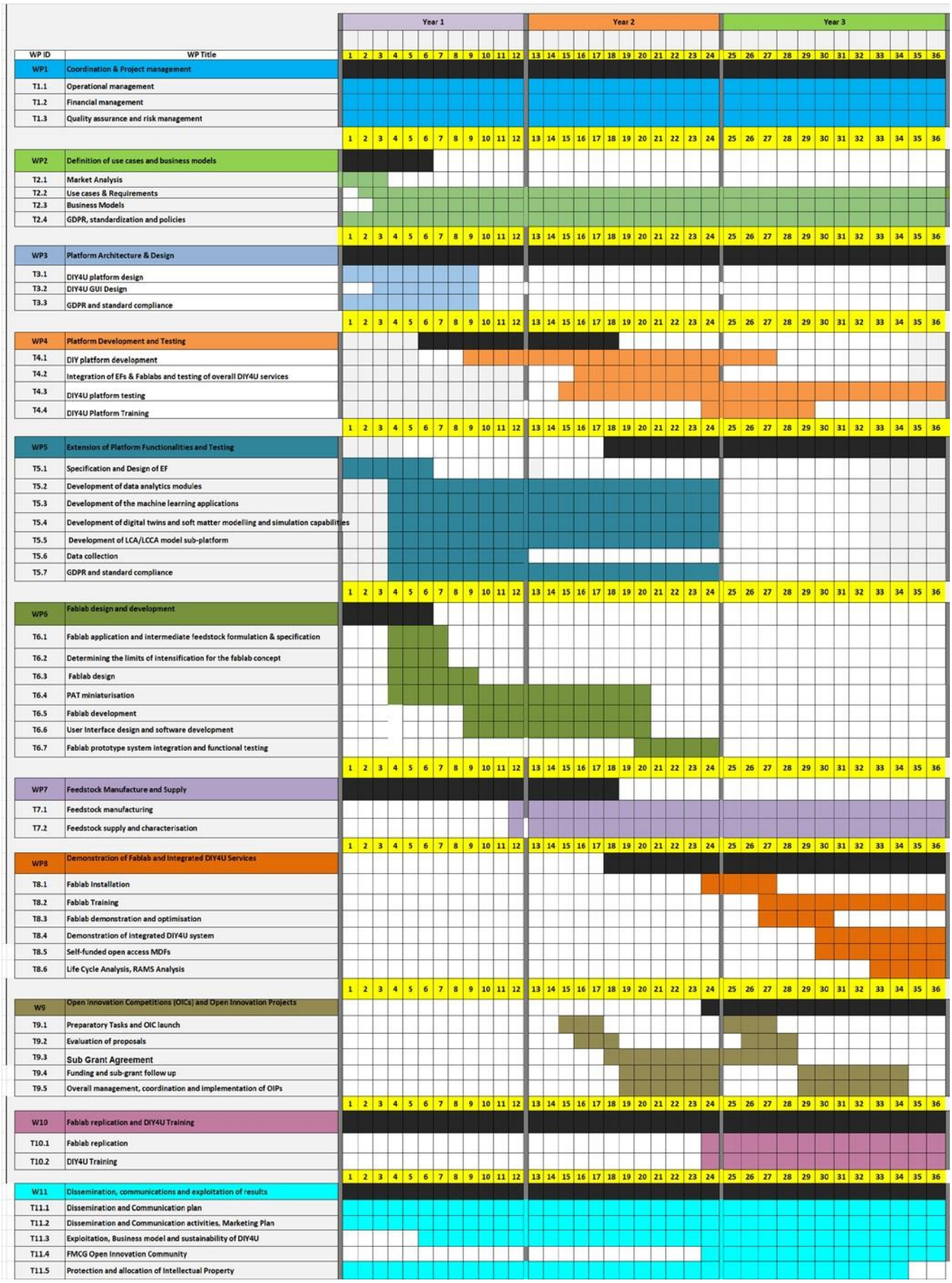


Figure 3.1a: The global strategy of the DIY4U project

DIY4U Gantt chart


Year 1
Year 2
Year 3
WP ID
WP Title
WP1 Coordination & Project management
T1.1 Operational management
T1.2 Financial management
T1.3 Quality assurance and risk management
WP2 Definition of use cases and business models
T2.1 Market Analysis
T2.2 Use cases & Requirements
T2.3 Business Models
T2.4 GDPR, standardization and policies
WP3 Platform Architecture & Design
T3.1 DIY4U platform design
T3.2 DIY4U GUI Design
T3.3 GDPR and standard compliance
WP4 Platform Development and Testing
T4.1 DIY platform development
T4.2 Integration of EFs & Fablabs and testing of overall DIY4U services
T4.3 DIY4U platform testing
T4.4 DIY4U Platform Training
WP5 Extension of Platform Functionalities and Testing
T5.1 Specification and Design of EF
T5.2 Development of data analytics modules
T5.3 Development of the machine learning applications
T5.4 Development of digital twins and soft matter modelling and simulation capabilities
T5.5 Development of LCA/LCCA model sub-platform
T5.6 Data collection
T5.7 GDPR and standard compliance
WP6 Fablab design and development
T6.1 Fablab application and intermediate feedstock formulation & specification
T6.2 Determining the limits of intensification for the fablab concept
T6.3 Fablab design
T6.4 PAT miniaturisation
T6.5 Fablab development
T6.6 User Interface design and software development
T6.7 Fablab prototype system integration and functional testing
WP7 Feedstock Manufacture and Supply
T7.1 Feedstock manufacturing
T7.2 Feedstock supply and characterisation
WP8 Demonstration of Fablab and Integrated DIY4U Services
T8.1 Fablab Installation
T8.2 Fablab Training
T8.3 Fablab demonstration and optimisation
T8.4 Demonstration of integrated DIY4U system
T8.5 Self-funded open access MDFs
T8.6 Life Cycle Analysis, RAMS Analysis
W9 Open Innovation Competitions (OICs) and Open Innovation Projects
T9.1 Preparatory Tasks and OIC launch
T9.2 Evaluation of proposals
T9.3 Sub Grant Agreement
T9.4 Funding and sub-grant follow up
T9.5 Overall management, coordination and implementation of OIPs
W10 Fablab replication and DIY4U Training
T10.1 Fablab replication
T10.2 DIY4U Training
W11 Dissemination, communications and exploitation of results
T11.1 Dissemination and Communication plan
T11.2 Dissemination and Communication activities, Marketing Plan
T11.3 Exploitation, Business model and sustainability of DIY4U
T11.4 FMCG Open Innovation Community
T11.5 Protection and allocation of Intellectual Property

3.2 Management structure and procedures

Organisational structure and the decision-making: We have chosen a clear and comprehensive project management structure that will enable us to efficiently monitor performance; manage risk and ensure that deliverables are achieved on time and within budget. The project will be governed by the Project Board comprised of a representative from each consortium member. The main decision-making responsibilities will lie within the participating partners ensuring they remain in control of the project direction and benefits. Each partner will have a vote and **decisions will be made by majority voting**. This will ensure that one partner cannot influence project decisions against the will of others. Figure 3.2 illustrates the DIY4U project management structure and reporting lines.

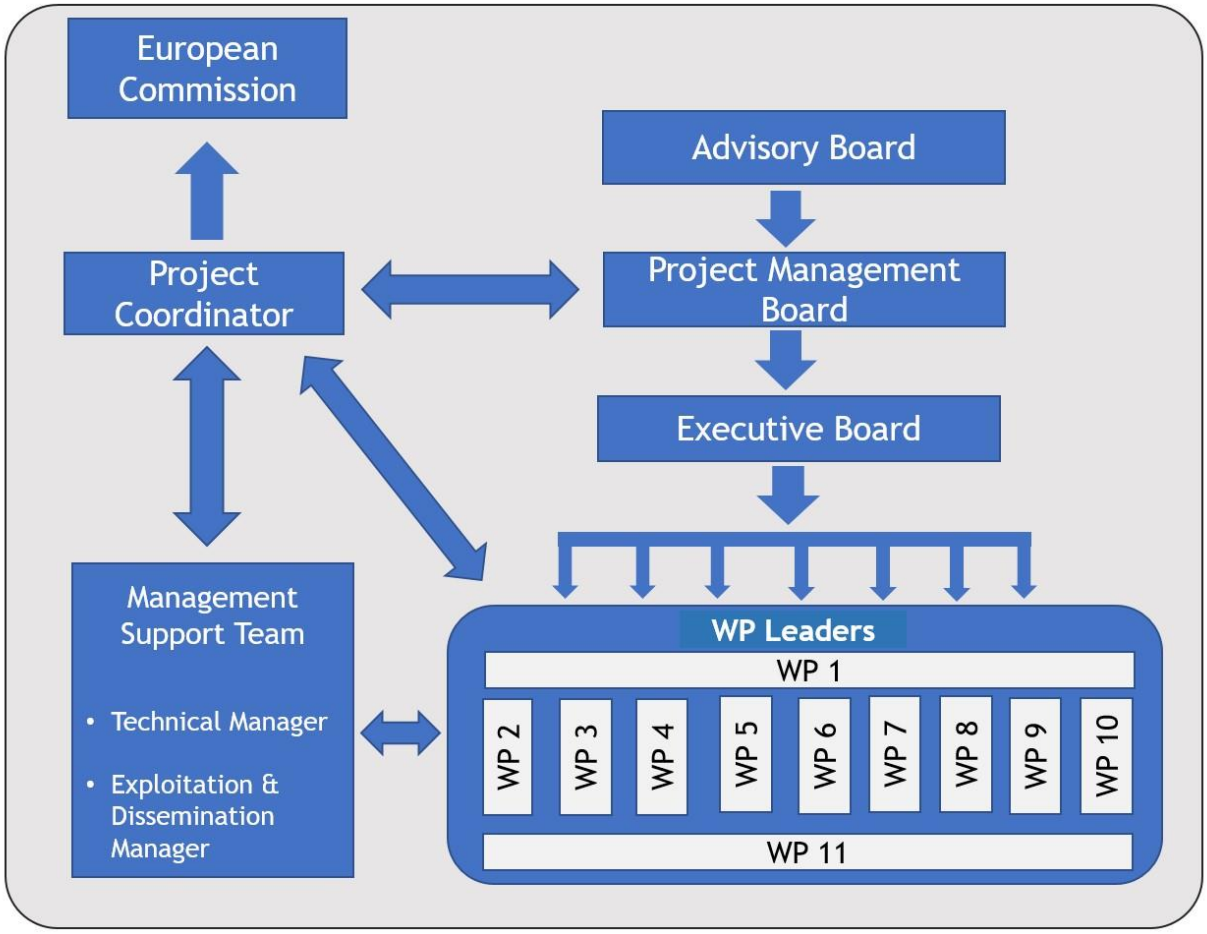


Figure 3.2 Management Structure and Reporting Lines

The responsibilities of each role shown in the project management structure (figure 3.2) is outlined below:

Project Management Board (PMB): The Project Management Board will comprise one delegate (Senior Representative) from each partner of the Consortium. It will be chaired by Dr Chandana Ratnayake as Project Coordinator. The PMB will be responsible for the overall success of the project and will therefore be the ultimate escalation point for any project issues. The PMB will approve the commencement of individual work packages based on the progress of specific milestones. The PMB will closely monitor milestones and critical stop/go decision points in accordance with the risk contingency plan. The PMB will also be solely responsible for resolving any administrative or contractual issues with the EC or within the consortium. The PMB will meet biannually (along with

project meetings when possible). Extraordinary PMB meetings will also be held when requested by the Project Manager.

Project Coordinator: The Project Coordinator will be the sole point of contact with the EC, and will oversee all project activities and collate the deliverables and milestone reports. He will also be responsible for: Legal, contractual, ethical, financial and administrative management of the consortium; Making sure the project consortium agreement is updated and managed when required; Organizing Project Steering Committee, Project Management, and Dissemination and Exploitation meetings; Collation of all costs statements and audit certificates and coordination of payments and distribution to relevant consortium members; Facilitating audits and overall liaison with the EC.
The DIY4U consortium has selected **Dr Chandana Ratnayake** from STEF to represent the consortium as Project Coordinator. He has over 15 years' experience in coordinating large projects including several European projects. His skills in team and strategy management and all other aspects of the technical project progression will be the key for this demanding project. He currently manages several large collaborative projects including H2020 at STEF. To appropriately manage the workload and allow the Project Coordinator to focus on the overall management of the project, we have created separate roles for the daily management of technical project issues.

Management Support Team: The Management Support Team comprises of the Technical Manager and Exploitation & Dissemination Manager, who's roles are defined below.

Project Technical Manager: The Project Technical Manager will be responsible for the direct management of technical project activities, monitoring technical progress and coordinating activities between Work Package Leaders. This will also involve maintaining the risk log and milestone progress reports forwarded to the Project Coordinator. The Project Technical Manager will therefore also be acting as Project Risk Manager. If a technical goal is to be dropped from the project the EC Scientific Officer will be informed of the recommendation or decision. Given that this role requires experience in international project management, we have elected **Dr Mark Taylor (Manager of High Throughput and Informatics at CPI)** for this role. Mark has 25 years' experience on working on development projects and delivery of work on a variety of innovative projects.

Exploitation and Dissemination Manager: The Dissemination, Exploitation and Communication Manager will report to the Project Coordinator and will be responsible for further developing the dissemination, exploitation and communication strategies. His responsibilities include: creation of the exploitation strategy, dissemination plan and communication policy; identification of future funding routes; approval of dissemination materials; identification of IPR and protection strategies. We have selected **Ken Andre Myhra Kihle (CEO at Cody)** for this role. Ken has 6 years of experience in bringing products to the market. He has expertise in IPR protection and exploiting research results. He will be closely supported by the IP and Marketing functions at CPI and other partner organisations.

Executive Board: Chaired by the Coordinator, the Executive Board (EB) will be responsible for the proper execution and implementation of the decisions of the Project Management Board. It will monitor the effective and efficient implementation of the Project. The EB members' appointment will be validated by the members of the PMB, and includes one representative per partner, including WP leaders. The EB will meet physically at least every 6 months. Additional meetings by teleconference will be organised when needed/requested.

WP Leaders: The WP Leaders are responsible for managing the Work Packages (WP) and all work conducted by participants in each WP. The WP Leaders establish, in coordination with the Project

Technical Manager, the detailed schedule of the WP and the work in progress. The Work Package leaders will be appointed within the organization leading each work packages.

Task Leaders: Task Leaders are responsible for managing the defined tasks within a WP, and all work conducted by participants in their tasks. The Task Leaders establish, in coordination with the WP Leader, the detailed schedule of the WP and the work in progress. Details of partners coordinating each task is included in the WP description. Task leaders will be appointed within the organization leading each task.

Advisory Board (AB): The Advisory Board will comprise leading experts from around the EU. They will be invited on the basis of their extensive knowledge of the systems and processes which will be impacted upon by the outcomes of DIY4U. Advisory Board members will be drawn from a wide range of related disciplines, thus covering all of the strategic guidance requirements of the project. Advisory Board members will not bear an executive responsibility but will be in regular contact with the Project Management Board and will meet with the Project Management Board annually. Members of the Advisory Board will receive briefings on the progress of the project and will make recommendations. Advisory Board members will be paid travel and subsistence costs to attend meetings of the project as required. We will select technical advisor from relevant FMCG manufacturer and ingredient suppliers, equipment manufacturers (e.g. HTP equipment, robotics, control system, etc) and FMCG retailers/supermarkets. The Project Coordinator will nominate Advisory Board members in collaboration with consortium partners. Each partner will be expected to suggest a relevant Advisory Board member.

To carter for the open innovation competitions, the following additional structures will be implemented:

Evaluation Committee Function: Committee responsible of selecting the best proposals in a 'Consensus Meeting'. Composition: It will be coordinated by PIC and composed by 'Project Board' plus 3 External Evaluators.
Selection Committee Function: Committee responsible of selecting the best proposals in the Jury Day and in the Review of Milestones during the open innovation program implementation. Composition: It will be coordinated by STEF and composed by 'Project Board' plus Advisory Board.
Mentoring Committee Function: Committee in charge of monitoring and validating the performance of the funded open innovation projects in the Interims and Final Review. Composition: It will be coordinated by
STEF/CPI and composed by the Technical Mentor and Business Mentor.
Ethical Committee Function: Committee in charge of monitoring and validating the ethical aspects of the proposals submitted. Composition: It will be coordinated by STEF and composed of the PMB and AB.

Effective Innovation Management: The management of knowledge and foreground IP created through the work carried out during the project will be the responsibility of the Exploitation Manager, Ken Andre Myhra Kihle. DIY4U will use a DESCA model consortium agreement. Within this consortium agreement the intended allocation of IPR will be defined which will be agreed by all participants in the project and will cover the IPR issues including ownership of knowledge, protection of that knowledge and access rights. Results generated in the project will be assessed in terms of IPR and whether necessary protection of the result is required. Further information on IP and IP management is described in section 2.2. We will commit the maximum effort to exploit the innovation outcome of the DIY4U project. WP11 has a dedicated task to develop a commercialization plan where we will carry out extensive research to find out the market opportunity and develop strategy to exploit it. For the successful outcome of the project we have included key liaisons with market users through the AB.

Risk Management and mitigation measure: As a project of novel technical objectives featuring complex and specialized technologies, DIY4U includes inherent risks towards delivering the expected results and achieving its goals. However, the project partners feel confident enough that these technical hazards can be well controlled and overcome, as a result of their expertise and the careful risk management approach. Each work package leader is required to maintain a risk log, which contains potential risks and a record of the occurred events. This information will be used to identify any issues and future risks that may arise.

3.3 Consortium as a whole

The consortium consists of 13 complementary members including research institutes, SMEs and a leading FMCG company. For effective management of the project and to listen to everyone's voice, formation of a high-level board consisting representative of each member of the consortium is a must. Hence, a Project Board is appropriate for such a big consortium. As this is a high-level board and senior officials from each partner organisation are involved, twice a year meeting is quite appropriate for this board. The creation of role of a *Project Coordinator*, who will chair the project board is very vital for successful implementation of such a project. The responsibility of this position is to coordinate between the consortium and the fund provider, EC. This position is important as the consortium must have to submit several reports regularly to the EC. They will also have to inform important decision taken in the project board meeting to the EC. For these regular and some critical jobs such a position is very crucial. Two other key management positions i.e. Project Technical Manager and Project Exploitation, Dissemination and Communication manager are very crucial as their roles have been discussed previously. Finally, the WP leaders' roles are vital as they will work in the grassroots level and they are responsible for delivering the actual deliverables of the project.

Partner	Role in the Project
STEF	SINTEF (Leading Norwegian RTO with extensive capabilities in soft matter processing and modelling & simulation) will be be the project coordinator for DIY4U project. STEF, along with CPI, will lead WP8 (Demonstration of Fablab and Integrated DIY4U Services), acting as MDF for the one of the two fablabs to be developed. STEF will provide crucial end-user input (WP2) that will feed into platform and fablab design/development (WPs 3-6) and support DSC and CTM in the development of relevant soft matter modelling and process simulation capabilities for DIY4U.
P&G	P&G (Leading global manufacturer of FMCGs including detergents and soaps) will guide the projects technical developments (WP3-WP8), ensuring that it remains industry-relevant and commercially viable. P&G will lead the manufacture and supply of base formulations (WP7) that will be customised in the MDFs (WP8). As a key exploitation partner, P&G will work with PIC and the other partners to develop the DIY4U business model and commercialization plan.
CPI	CPI (Part of the UK's High Value Manufacturing Catapult, hosting the National Formulation Centre with capabilities in the processing of complex particle, complex liquids and other material formulations, as well as capabilities in modelling and process engineering) will, similarly to STEF, lead WP8 (Demonstration of Fablab and Integrated DIY4U Services), acting as MDF for the one of the two fablabs to be developed, and provide crucial end-user input (WP2) that will feed into platform and fablab design/development (WPs 3-6). CPI will additionally employ a combination of its formulation, process engineering and modelling & simulation capabilities to lead the development of the LCA/LCCA model subplatform (WP4) and support DSC and CTM in the development of soft matter modelling & simulation capabilities and digital twin.

VTT	VTT (Leading European RTO with PAT, data analytics and automation expertise and capabilities) will act as the leader of PAT activities in the project, developing integrated, miniaturised and lowcost PAT methods and devices for the Fablabs based on near-infrared spectroscopy, camera-based measurements and other techniques. VTT will also develop calibration methods and data analysis for the PAT devices.
CTM	Cetemet (robotic and automation expert, with process modelling expertise) will co-lead the extension of platform functionalities and testing, developing a digital twin and deploying a soft matter modelling and simulation for using it to provide an open access collaborative production. Cetemet will enhance the simulation of soft matter by simulating it at the macroscopic level and correlate the results obtained at a microscopic level and analysing its interaction with the equipment that contains it. For this, it will carry out flow simulations with fluid particles to analyze their behavior and interaction with the structure through which the flow circulates by means of structural and fluid dynamics simulations with finite elements. The results will be implemented in the virtual twin in the platform and will contribute to the testing of the same. In addition, Cetemet will support the design and development of the fablab prototypes, using 3D CAD systems to obtain 3D models and drawings according to the requirements that will contribute to the development of the system. This activity will also involve the use of FE simulations for the assurance of the material resistance and the structural stability of the prototype.
IRIS	IRIS will function as the main software developer of the consortium, responsible for the programming and deployment of the KMC platform which will act as core controller and provide the backend and frontend for the whole DIY4U platform. IRIS will therefore be also responsible for the integration of the various advanced tools in the form of modules of the platform. IRIS will collaborate closely with EFF and DCC and the partners developing the extension functionalities/modules as well as fablab users (STEF and CPI) and the rest of the consortium.
DCC	Digital Catapult, as the blockchain expert of the consortium, will contribute to the understanding of user needs and requirements to then lead on the platform architecture design, having into account both the centralised and decentralised elements of the solution to enable efficient and trusted relationships between the different stakeholders involved in the fabrication of detergents. DCC will also have an advisory role during the development process of the platform, making sure the design proposed is being met. Moreover, DCC will support the dissemination and exploitation actions (specially with the blockchain and manufacturing communities) and will run the FMCG Open Innovation Community with the purpose to bring relevant stakeholders together and find research and business opportunities. Moreover, DCC will run workshops to identify new business models emerging for start-ups and collaborators and will produce reports from the results for knowledge sharing and to raise awareness of DIY4U.
PIC	PICOM (French innovation cluster working for the retail industry) will lead use case definition and business model definition as well as evaluation, ensuring that they address both customers, brands, retailers, services providers and technology providers' needs. PICOM will also provide strong support
	to open calls management and dissemination.
CDY	Cody (robotic and automation solution provider with expertise in manufacturing automated equipment) will lead the Fablab design and development. Cody will contribute with their knowledge within machine development, product development, machine safety, operator interface and interface between machines and platform. Cody will also lead the Fablab replication and DIY4U training.
RDI	RDI will analyse Market and segment sectors related to the DIY4U. Ensure data repository and analytics in order to valuable information and non-trivial correlation between parameters, assess the human behaviour and needs. Also, RDI will create Machine Learning to evaluate created formulas and make decisions.

STL	Stelar ("Privacy-by-Design" experts with legal background) will contribute legal data privacy requirements analysis and GDPR/other regulations compliance support monitoring, ensuring that DIY4U complies with relevant regulations.
EFF	Effective Decisions, actively involved in the blockchain space in the last 4 years, will contribute with high-level architecture perspective and development expertise for the decentralized modules, distributed ledger technologies and privacy preserving features of the DIY4U platform.
DSC	DSC (soft matter modelling and process simulation experts) will lead the development of relevant soft matter models and support CTM with the development of the Fablab digital Twin, working in close collaboration with STEF, CPI and the rest of the consortium.